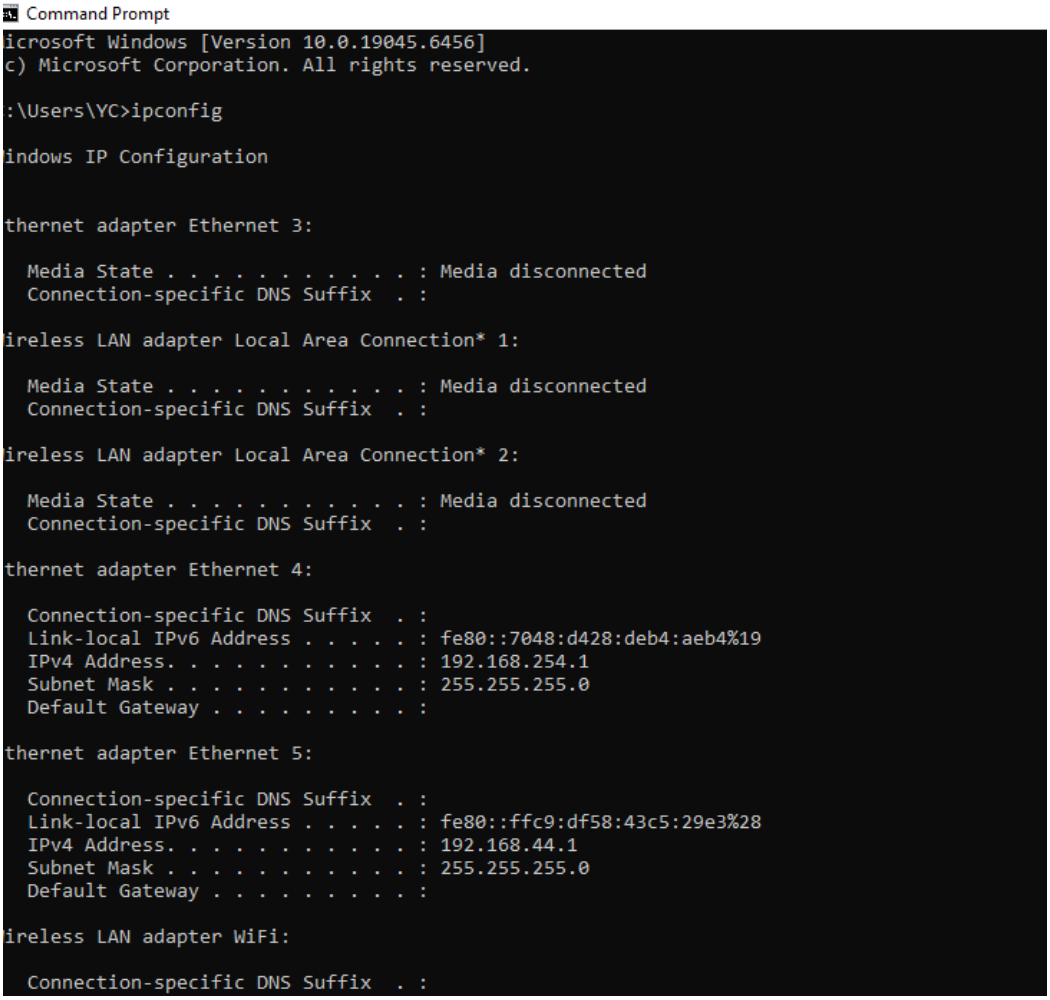
Cyber Security labs

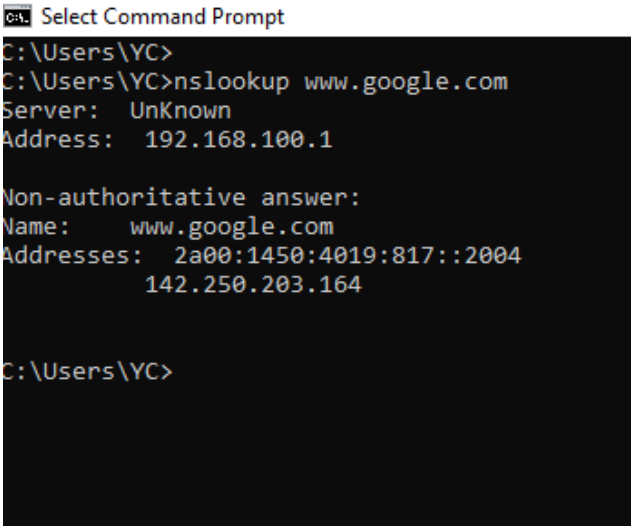
Wednesday, October 22, 2025 7:37 AM

Lab1 : IP check through command prompt

- Apna local ip address check krny k leye comand ipconfig



- Kisi website ka ip check krny k leye command nslookup



- Apna public internet ip address check krny k leye comamnd nslookup myip.opendns.com resolver1.opendns.com

```
C:\Users\YC>nslookup myip.opendns.com resolver1.opendns.com
Server:  dns.sse.cisco.com
Address:  208.67.222.222

Non-authoritative answer:
Name:    myip.opendns.com
Address:  103.150.154.162


C:\Users\YC>
```

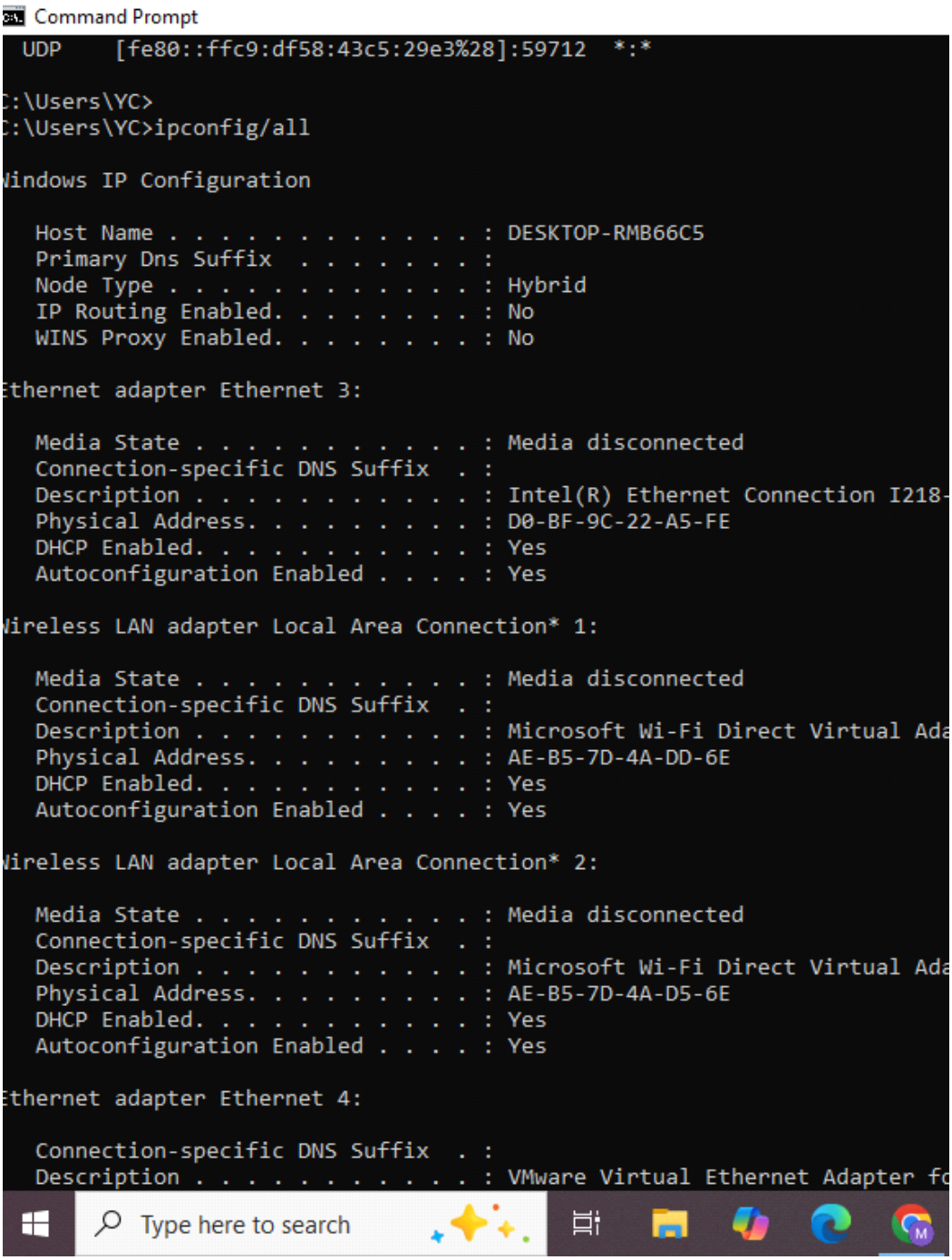
- Network connection details check krny k leye command netstat -an

```
C:\Users\YC>netstat -an

Active Connections

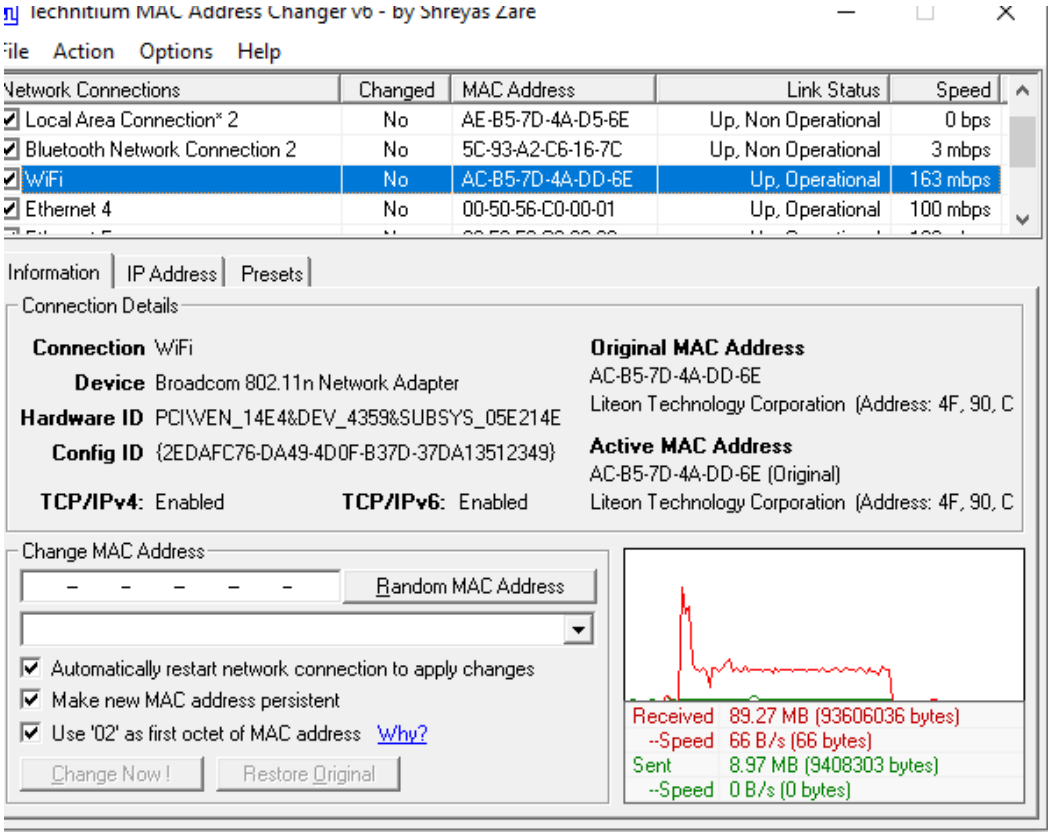
  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:902            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:912            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:3306           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5357           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:7680           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:33060          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49669          0.0.0.0:0              LISTENING
  TCP    127.0.0.1:1001         0.0.0.0:0              LISTENING
  TCP    127.0.0.1:49673        127.0.0.1:49674        ESTABLISHED
  TCP    127.0.0.1:49674        127.0.0.1:49673        ESTABLISHED
  TCP    127.0.0.1:49675        127.0.0.1:49676        ESTABLISHED
  TCP    127.0.0.1:49676        127.0.0.1:49675        ESTABLISHED
  TCP    127.0.0.1:64219        0.0.0.0:0              LISTENING
  TCP    127.0.0.1:64219        127.0.0.1:64408        ESTABLISHED
  TCP    127.0.0.1:64408        127.0.0.1:64219        ESTABLISHED
  TCP    192.168.44.1:139       0.0.0.0:0              LISTENING
  TCP    192.168.100.6:139      0.0.0.0:0              LISTENING
```

- Apny pc ka mac address check krny k leye command ipconfig/all

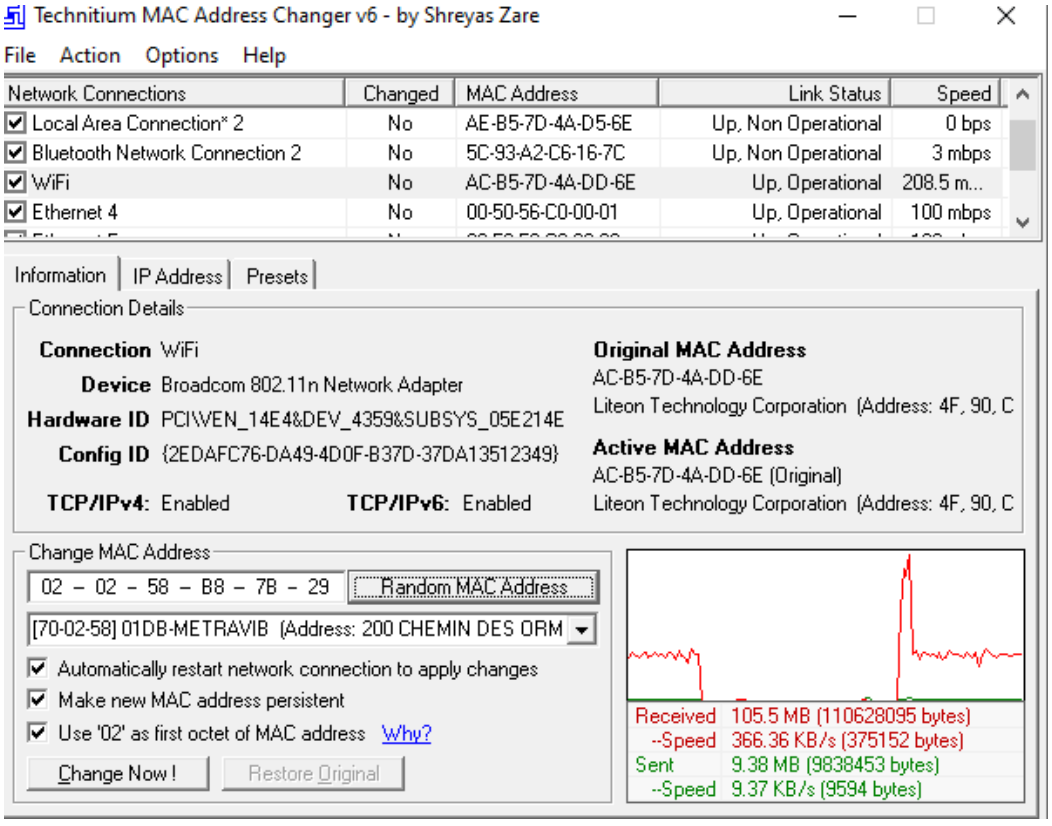


Lab2: change mac address with TMAC

- Select any one from network connection



- Click on random button



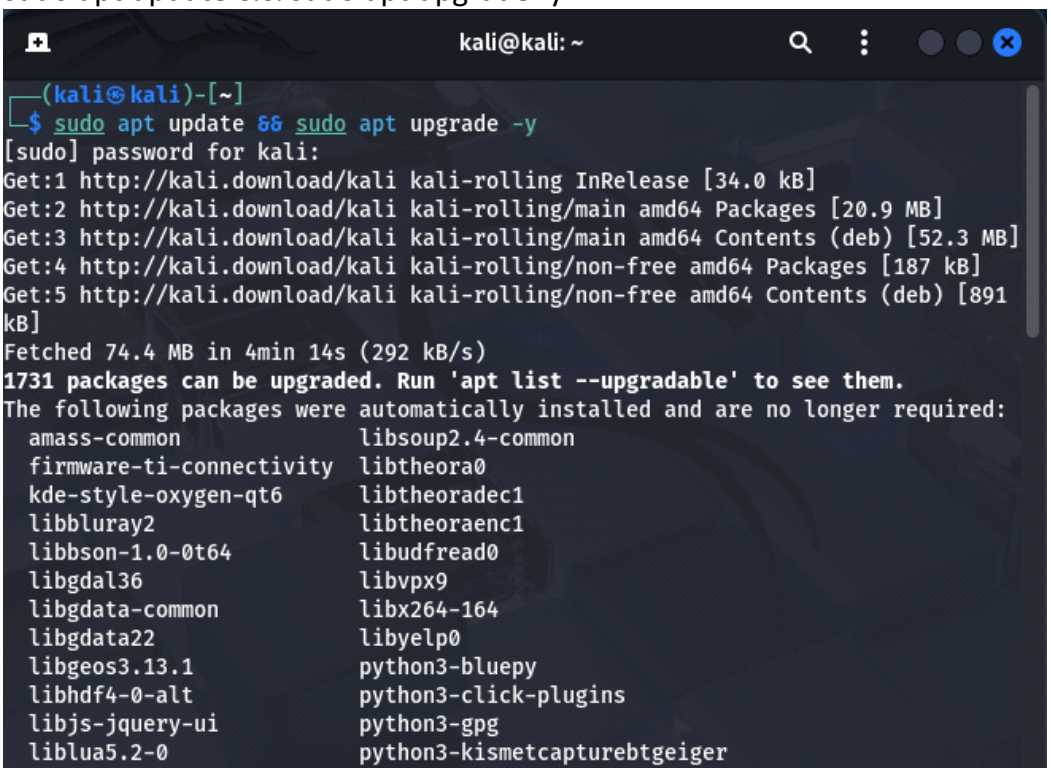
- Click on **change now!**
- To restore orrignal mac address click the button **restore original**.

Iab3: Installation of kali

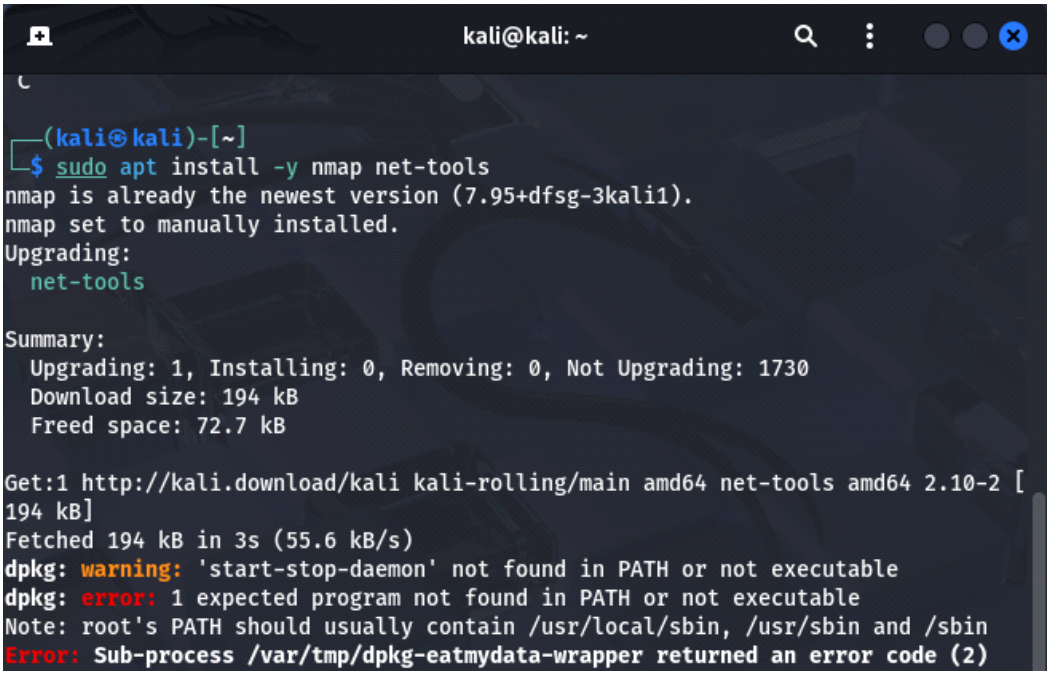
- Download wmware workstation
- Install kali linux IOS (file type : Installer IOS 64-bit)
- Open vmware and click new virtual machine
- Select installer disk image IOS file
- Browse --> select your **kali IOS file** --> next
- Select guest OS linux verion Dabain 12 × 64 - bit
- Put virtual machine name and loacation (where kali ios download)
- Select disk size maximum 4gb
- Select “Store virtual disk as a single file”
- Click **Next** → **Finish**
- Edit VM setting (optional) memory 4gb , processors 2 minimum
- Set **Network Adapter**
- Click **next**
- Start installation. Click **Power on this virtual machine**.
- choose **Graphical Install**
- Choose Language English
- User setup (host name = kali) ,(user name = kali) ,password(kali)
- Partition disks :- select **Guided – use entire disk**
- Select the default virtual disk (40 GB)
- Partitioning scheme: **All files in one partition**
- Finish partitioning → Write changes to disk → **Yes**

Lab4 : Assign IP on kali linux

- Command for update and install tools
  - sudo apt update && sudo apt upgrade -y



  - Sudo apt install nmap -y nmap net-tools

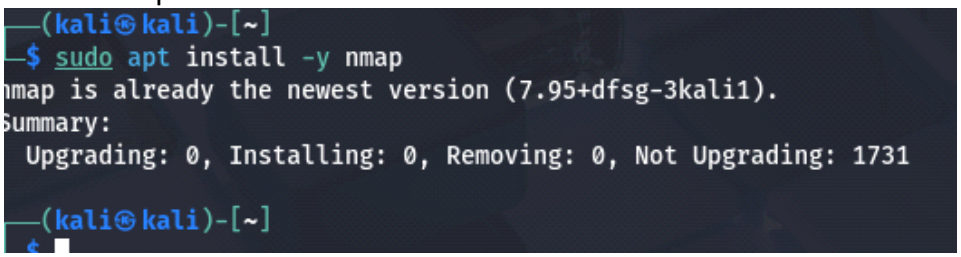


- To assign ip



Lab 5 : Nmap

- Install nmap



- Verify nmap version

```
┌──(kali㉿kali)-[~]
└─$ nmap -version
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.0 libssh2-1.11.1 libz-1.3.1 libpcre2-10.
45 libpcap-1.10.5 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select

┌──(kali㉿kali)-[~]
└─$
```

- Ports scan on host

```
┌──(kali㉿kali)-[~]
└─$ nmap --top-ports 100 150.1.7.100
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-22 23:55 EDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 4.19 seconds
```

- Deeply scan

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -sV -A 150.1.7.100
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-23 00:02 EDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 72.59 seconds

┌──(kali㉿kali)-[~]
└─$
```

- UDP scan
  - Nmap -sU 150.1.7.0/24

```
┌──(kali㉿kali)-[~]
└─$ nmap -sU 150.1.7.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-25 23:57 EDT
Nmap scan report for 150.1.7.0
Host is up (0.00047s latency).
All 1000 scanned ports on 150.1.7.0 are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap scan report for 150.1.7.1
Host is up (0.00036s latency).
All 1000 scanned ports on 150.1.7.1 are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap scan report for 150.1.7.2
Host is up (0.00066s latency).
All 1000 scanned ports on 150.1.7.2 are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap scan report for 150.1.7.3
Host is up (0.0011s latency).
All 1000 scanned ports on 150.1.7.3 are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap scan report for 150.1.7.4
```

  - Single
    Nmap 150.1.7.102/24

```
┌──(kali㉿kali)-[~]
└─$ nmap 150.1.7.102/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 00:39 EDT
Nmap done: 256 IP addresses (0 hosts up) scanned in 217.51 seconds

┌──(kali㉿kali)-[~]
└─$
```

- Range

  Nmap 150.1.7.102-110

```
┌──(kali㉿kali)-[~]
└─$ nmap 150.1.7.102-110
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 00:45 EDT
Nmap done: 9 IP addresses (0 hosts up) scanned in 9.19 seconds

┌──(kali㉿kali)-[~]
└─$
```

- Subnet scan

  Nmap 150.1.7.102/24

```
┌──(kali㉿kali)-[~]
└─$ nmap 150.1.7.102/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 00:39 EDT
Nmap done: 256 IP addresses (0 hosts up) scanned in 217.51 seconds

┌──(kali㉿kali)-[~]
└─$
```

- Aggressive scan

  Nmap -sV -A -T4 150.1.7.102

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV -A -T4 150.1.7.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 00:50 EDT
Nmap scan report for 150.1.7.102
Host is up (0.0080s latency).
All 1000 scanned ports on 150.1.7.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 o
pen and 1 closed port
Aggressive OS guesses: D-Link DFL-700 firewall (89%), HP Officejet Pro 8500 prin
ter (89%), IBM i 7.4 (89%), ReactOS 0.3.7 (89%), Sanyo PLC-XU88 digital video pr
ojector (89%), Sonus GSX9000 VoIP proxy (88%), Asus WL-500gP wireless broadband
router (88%), Microsoft Windows 2000 (88%), Microsoft Windows Server 2003 Enterp
rise Edition SP2 (88%), Microsoft Windows Server 2003 SP2 (88%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 2 hops

TRACEROUTE (using port 80/tcp)
```

- Single port scan

  Nmap -p 80 150.1.7.102

```
┌──(kali㉿kali)-[~]
└─$ nmap -p 80 150.1.7.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 00:52 EDT
Nmap scan report for 150.1.7.102
Host is up (0.0015s latency).

PORT   STATE    SERVICE
80/tcp filtered http

Nmap done: 1 IP address (1 host up) scanned in 0.51 seconds
```

- Range port scan

Nmap -p 1-500 150.1.7.102

```
┌──(kali㉿kali)-[~]
└─$ nmap -p 1-500 150.1.7.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 00:53 EDT
Nmap scan report for 150.1.7.102
Host is up (0.00063s latency).
All 500 scanned ports on 150.1.7.102 are in ignored states.
Not shown: 500 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 3.03 seconds

┌──(kali㉿kali)-[~]
└─$
```

- Fast scan
  Nmap -F 150.1.7.102 (scan 100 ports)

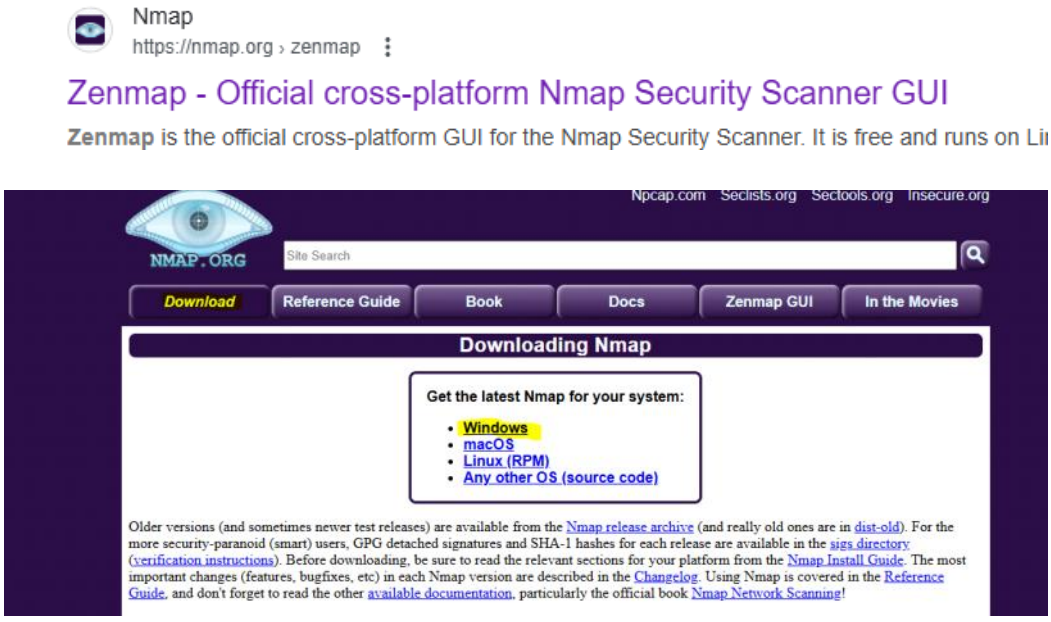
```
┌──(kali㉿kali)-[~]
└─$ nmap -F 150.1.7.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 00:54 EDT
Nmap scan report for 150.1.7.102
Host is up (0.00083s latency).
All 100 scanned ports on 150.1.7.102 are in ignored states.
Not shown: 100 filtered tcp ports (no-response)

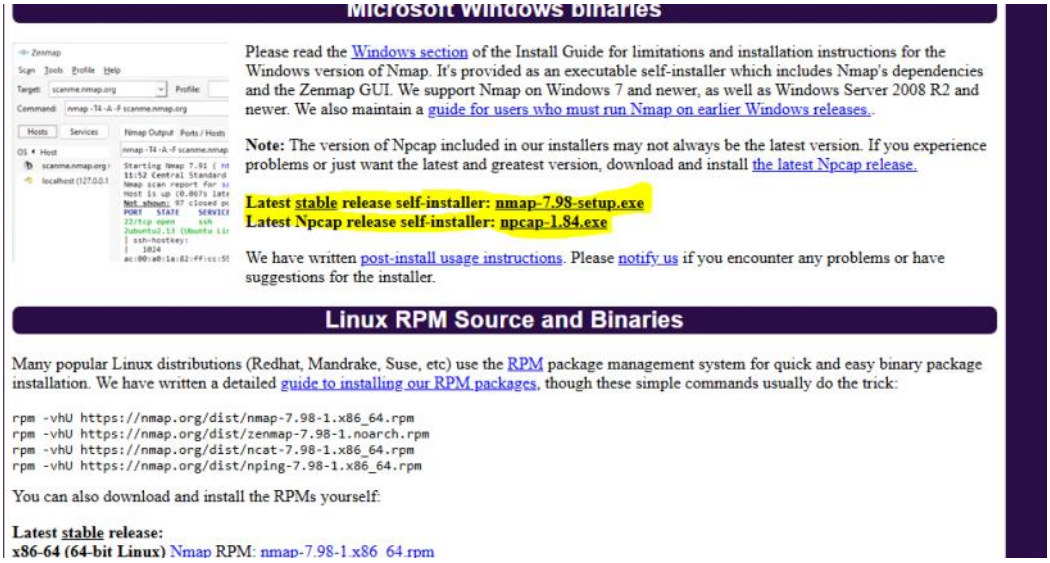
Nmap done: 1 IP address (1 host up) scanned in 2.03 seconds

┌──(kali㉿kali)-[~]
└─$
```

- All ports scan
  Nmap -p -p -65535
- All ports scan
  Nmap -p -150.1.7.102

- Zenmap
  - Download zenmap from google





Microsoft Windows binaries

Please read the Windows section of the Install Guide for limitations and installation instructions for the Windows version of Nmap. It's provided as an executable self-installer which includes Nmap's dependencies and the Zenmap GUI. We support Nmap on Windows 7 and newer, as well as Windows Server 2008 R2 and newer. We also maintain a guide for users who must run Nmap on earlier Windows releases..

Note: The version of Npcap included in our installers may not always be the latest version. If you experience problems or just want the latest and greatest version, download and install the latest Npcap release.

Latest stable release self-installer: nmap-7.98-setup.exe
Latest Npcap release self-installer: npcap-1.84.exe

We have written post-install usage instructions. Please notify us if you encounter any problems or have suggestions for the installer.

Linux RPM Source and Binaries

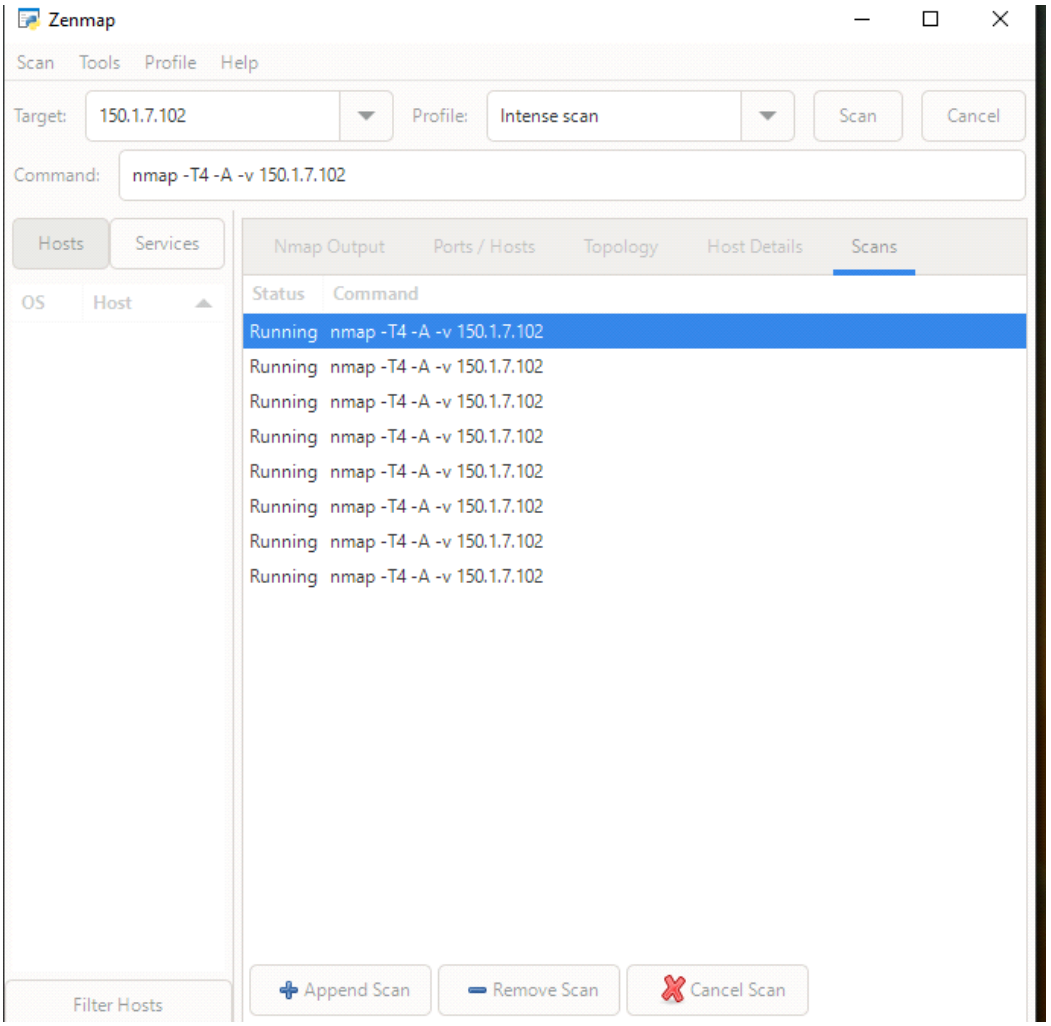
Many popular Linux distributions (Redhat, Mandrake, Suse, etc) use the RPM package management system for quick and easy binary package installation. We have written a detailed guide to installing our RPM packages, though these simple commands usually do the trick:

```
rpm -vhU https://nmap.org/dist/nmap-7.98-1.x86_64.rpm
rpm -vhU https://nmap.org/dist/zenmap-7.98-1.noarch.rpm
rpm -vhU https://nmap.org/dist/ncat-7.98-1.x86_64.rpm
rpm -vhU https://nmap.org/dist/nping-7.98-1.x86_64.rpm
```

You can also download and install the RPMs yourself:

Latest stable release:
x86-64 (64-bit Linux) Nmap RPM: nmap-7.98-1.x86_64.rpm

- Run the zenmap as administrator
- Target : 150.1.7.102
- Profile: intense scan
- Click on scan button



Lab 6: Subdomain finder in kali linux through different websites

- Subdomain finder
  - sudo apt install assetfinder

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install assetfinder
Installing:
  assetfinder

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1735
  Download size: 1,980 kB
  Space needed: 5,978 kB / 58.1 GB available

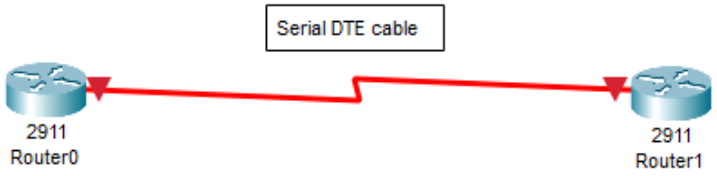
Get:1 http://http.kali.org/kali kali-rolling/main amd64 assetfinder amd64 0.1.1-
1+b10 [1,980 kB]
Fetched 1,980 kB in 12s (164 kB/s)
dpkg: warning: 'start-stop-daemon' not found in PATH or not executable
dpkg: error: 1 expected program not found in PATH or not executable
Note: root's PATH should usually contain /usr/local/sbin, /usr/sbin and /sbin
Error: Timeout was reached
Error: Sub-process /var/tmp/dpkg-eatmydata-wrapper returned an error code (2)

┌──(kali㉿kali)-[~]
└─$
```

Lab7: Cisco Packet Tracer

- Assign IP to the routers and PCs and end-to-end connectivity

  Step 1 : connect 2 routers and assign them ip



  - IP for router0

```
rl>en
rl#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
rl(config)#hostname Rl
Rl(config)#int s 0/2/0
Rl(config-if)#ip add 2.1.1.1 255.0.0.0
Rl(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
Rl(config-if)#exit
Rl(config)#exit
Rl#
%SYS-5-CONFIG_I: Configured from console by console

Rl#show ip int brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively
down down
GigabitEthernet0/1     unassigned      YES unset  administratively
down down
GigabitEthernet0/2     unassigned      YES unset  administratively
down down
Serial0/2/0            2.1.1.1         YES manual down
```

Ctrl+F6 to exit CLI focus Copy Paste

```
down down
Serial0/2/0            2.1.1.1         YES manual down
down
Serial0/2/1            unassigned      YES unset  administratively
down down
Vlan1                  unassigned      YES unset  administratively
down down
Rl#en
Rl#en
Rl#exit




Rl con0 is now available
```

- IP for router1

```
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname r2
r2(config)#int s 0/2/0
r2(config-if)#ip add 2.1.1.2 255.0.0.0
r2(config-if)#no shut

r2(config-if)#
%LINK-5-CHANGED: Interface Serial0/2/0, changed state to up

r2(config-if)#exit
r2(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed
state to up
exit
r2#
%SYS-5-CONFIG_I: Configured from console by console

r2#show ip int brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively
```

IOS Command Line Interface

```
r2(config-if)#exit
r2(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed
state to up
exit
r2#
%SYS-5-CONFIG_I: Configured from console by console

r2#show ip int brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively
down down
GigabitEthernet0/1     unassigned      YES unset  administratively
down down
GigabitEthernet0/2     unassigned      YES unset  administratively
down down
Serial0/2/0            2.1.1.2         YES manual up
up
Serial0/2/1            unassigned      YES unset  administratively
down down
Vlan1                  unassigned      YES unset  administratively
down down
r2#
```

Ctrl+F6 to exit CLI focus

Copy Paste

```
Serial0/2/1            unassigned      YES unset  administratively
down down
Vlan1                  unassigned      YES unset  administratively
down down
r2#
r2#exit

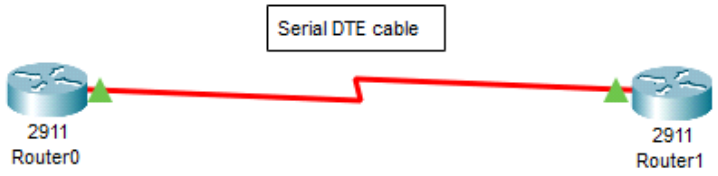


r2 con0 is now available

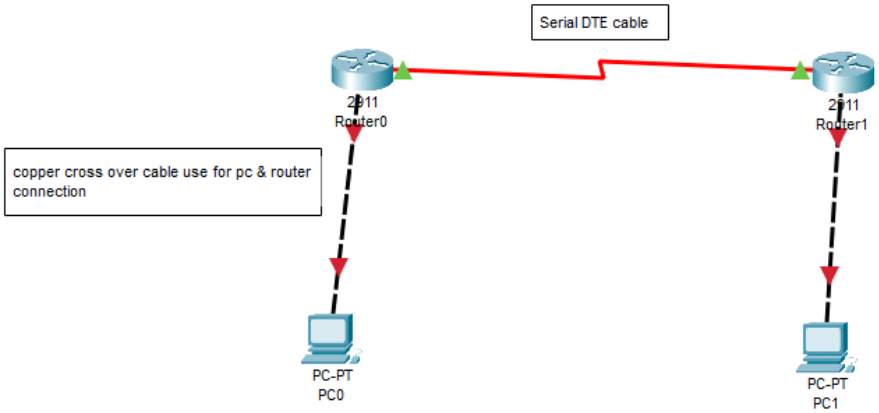


Press RETURN to get started.
```

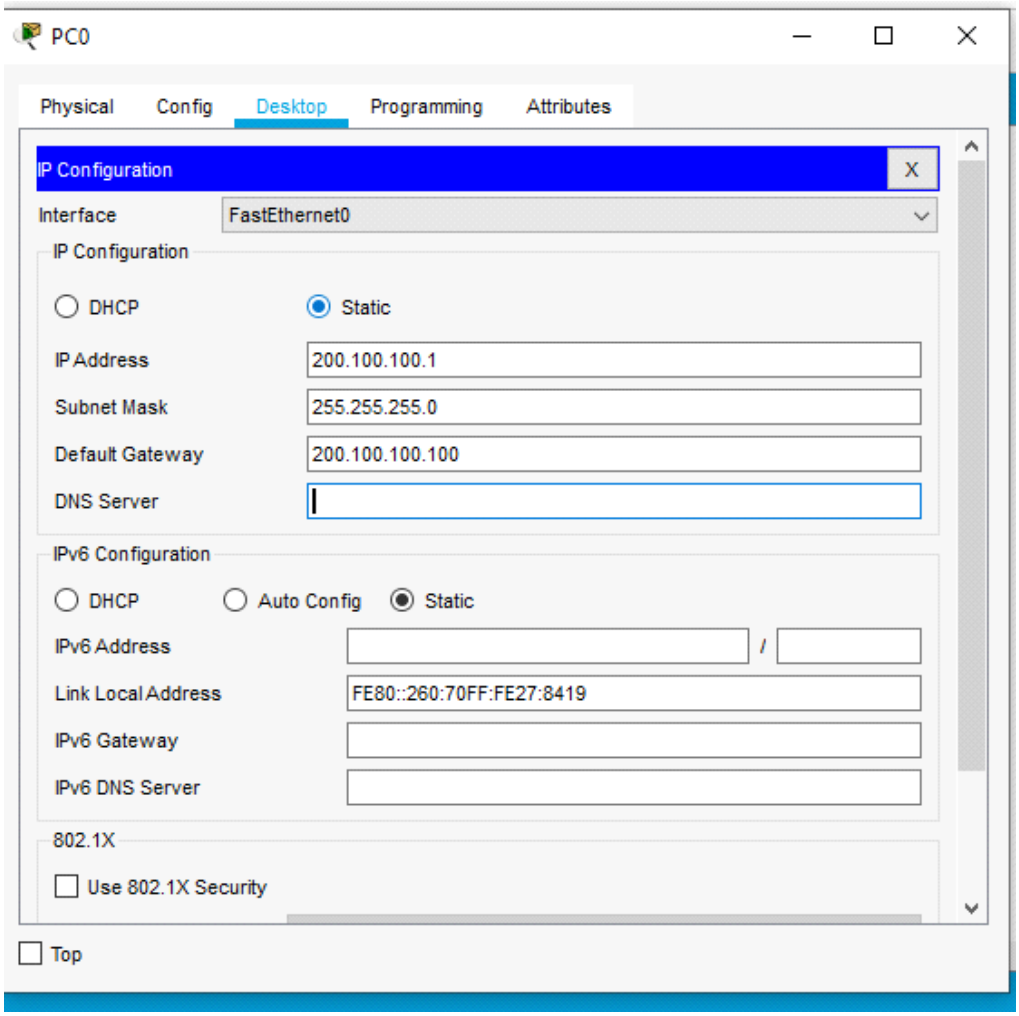
➢ Connection establish b/w routers



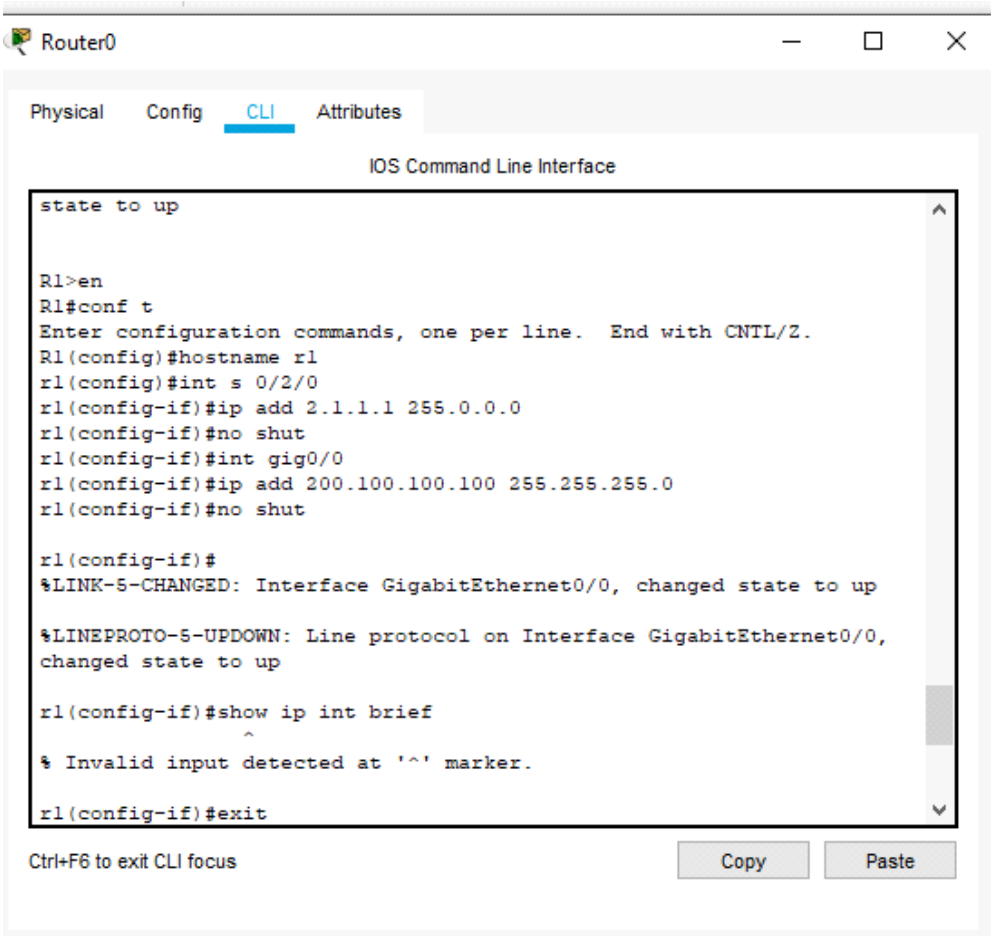
- Step 2 : Connect pc's with routers

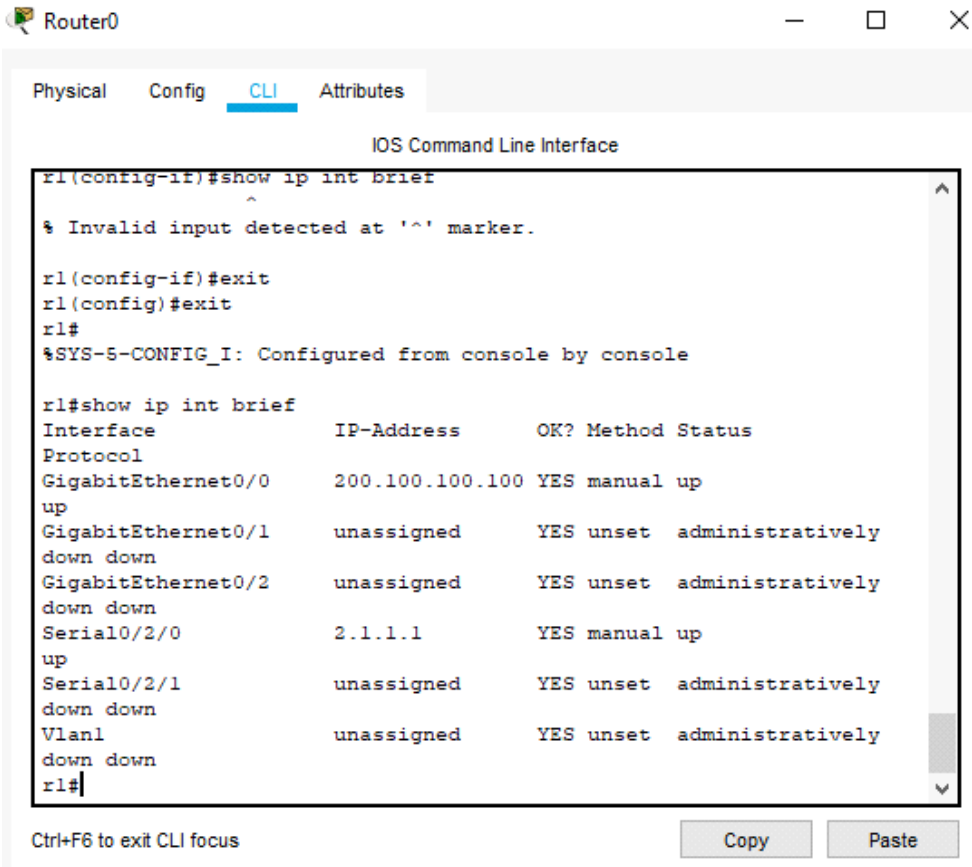


➢ IP for PC0



- Go to router0





- Open command prompt on PC0



```
C:\>
C:\>ping 200.100.100.100

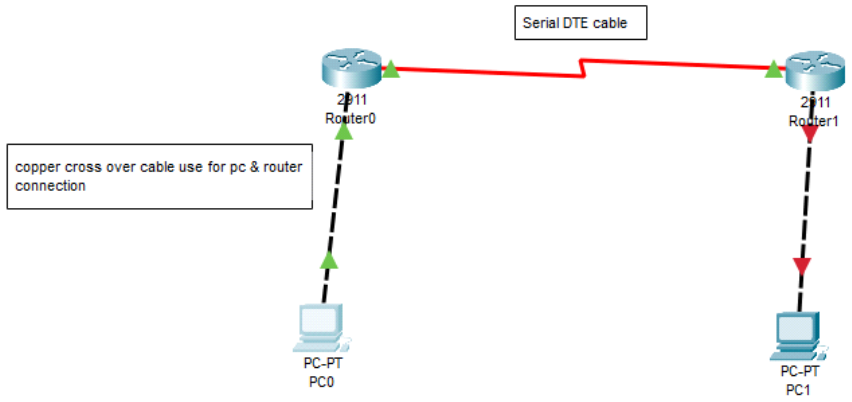
Pinging 200.100.100.100 with 32 bytes of data:

Reply from 200.100.100.100: bytes=32 time=1ms TTL=255
Reply from 200.100.100.100: bytes=32 time<1ms TTL=255
Reply from 200.100.100.100: bytes=32 time<1ms TTL=255
Reply from 200.100.100.100: bytes=32 time<1ms TTL=255

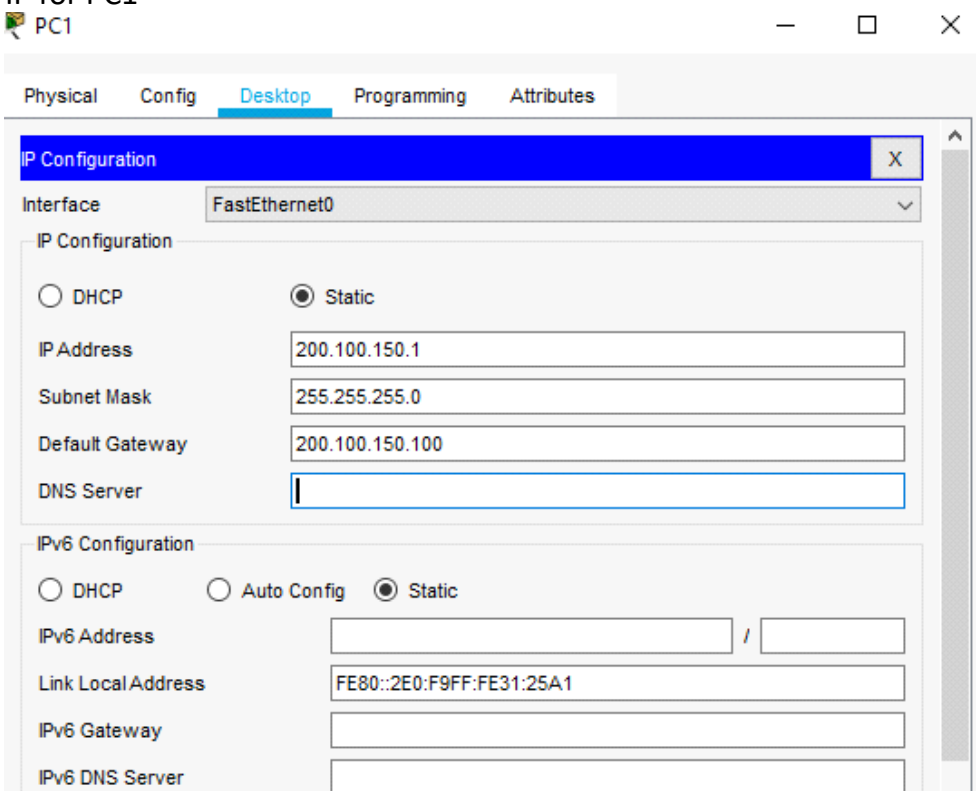
Ping statistics for 200.100.100.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

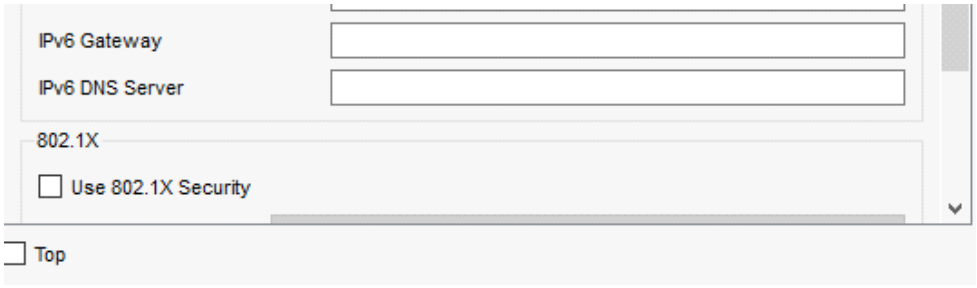
C:\>
```

- Connection b/w router0 and PC0

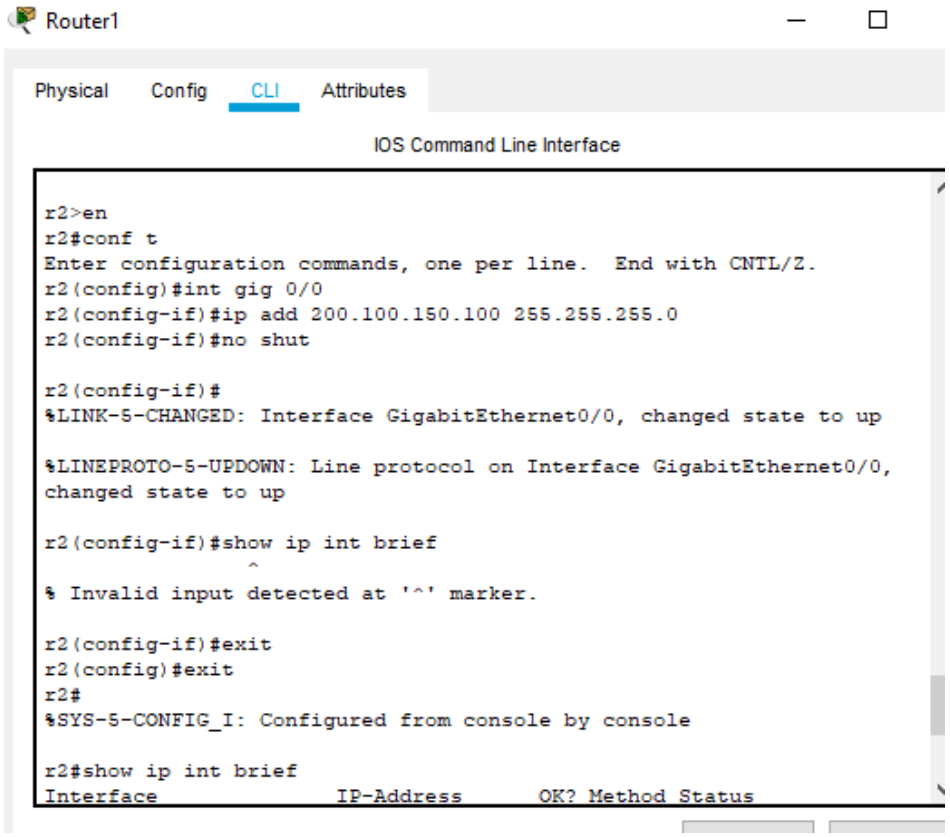


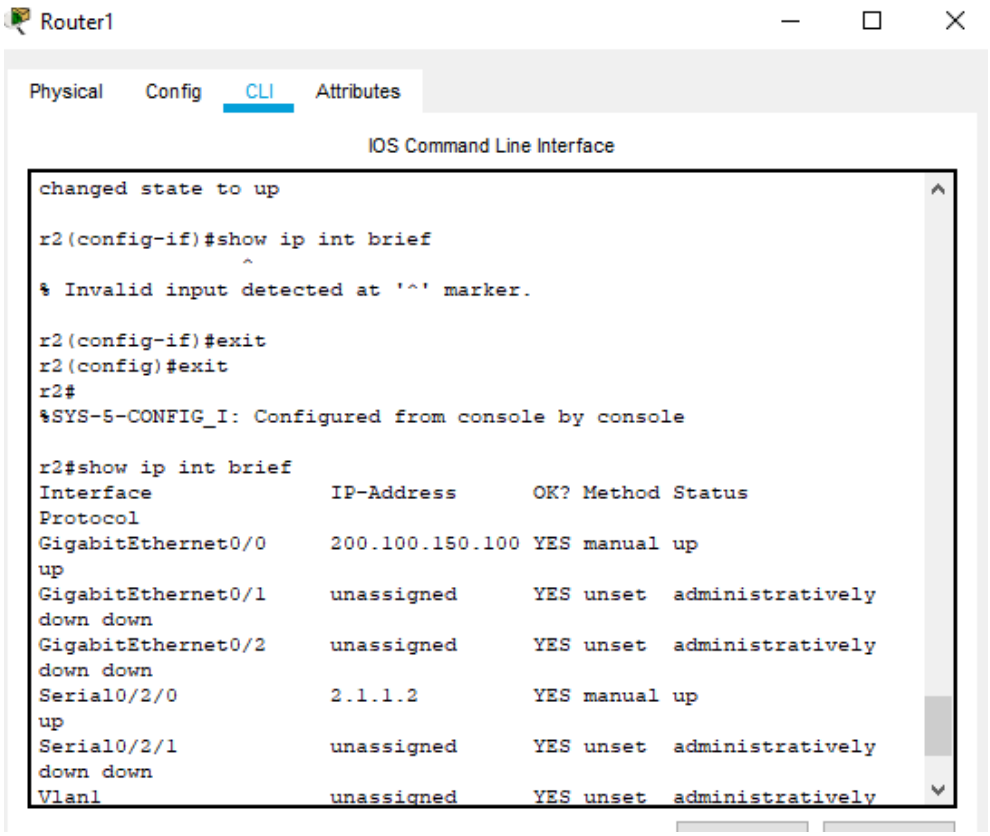
- IP for PC1



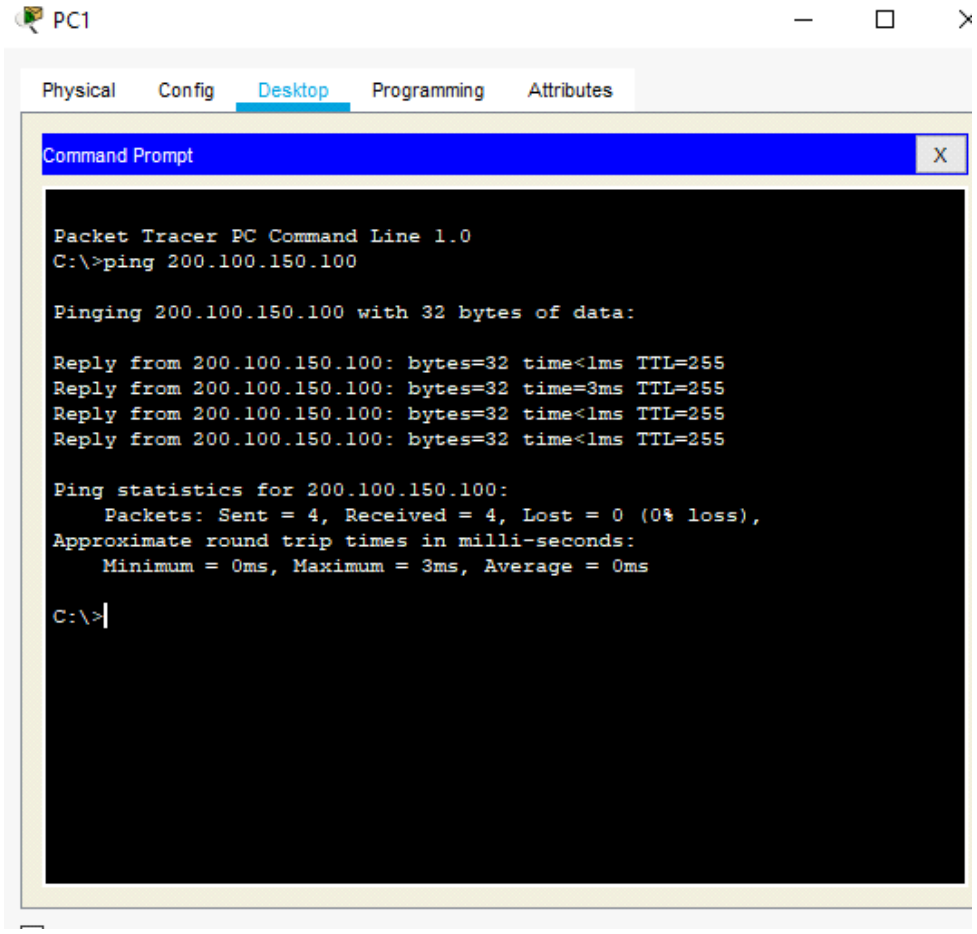


- Go to router1





- Open command prompt on PC1



- Router1 to PC0 ping

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
r2>
r2>en
r2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r2(config)#ping 200.100.100.1
           ^
% Invalid input detected at '^' marker.

r2(config)#ip route 200.100.100.0 255.255.255.0 2.1.1.1
r2(config)#exit
r2#
%SYS-5-CONFIG_I: Configured from console by console

r2#
r2#ping 200.100.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.100.100.1, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/8 ms

r2#
```

Ctrl+F6 to exit CLI focus Copy Paste

- Ping PC0 to PC1

```
C:\>ping 200.100.150.1

Pinging 200.100.150.1 with 32 bytes of data:

Reply from 200.100.100.100: Destination host unreachable.
Reply from 200.100.100.100: Destination host unreachable.
Reply from 200.100.100.100: Destination host unreachable.
Reply from 200.100.100.100: Destination host unreachable.

Ping statistics for 200.100.150.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

- Router0 to PC1 ping

```
rl>en
rl#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
rl(config)#ip route 200.100.150.0 255.255.255.0 2.1.1.2
rl(config)#exit
rl#
%SYS-5-CONFIG_I: Configured from console by console

rl#ping 200.100.100.0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.100.100.0, timeout is 2
seconds:

Reply to request 0 from 200.100.100.1, 1 ms
Reply to request 1 from 200.100.100.1, 3 ms
Reply to request 2 from 200.100.100.1, 3 ms
Reply to request 3 from 200.100.100.1, 11 ms
Reply to request 4 from 200.100.100.1, 0 ms
```

```
Reply to request 2 from 200.100.100.1, 3 ms
Reply to request 3 from 200.100.100.1, 11 ms
Reply to request 4 from 200.100.100.1, 0 ms

r1#
```

- Ping PC1 to PC0

```
C:\>ping 200.100.100.0

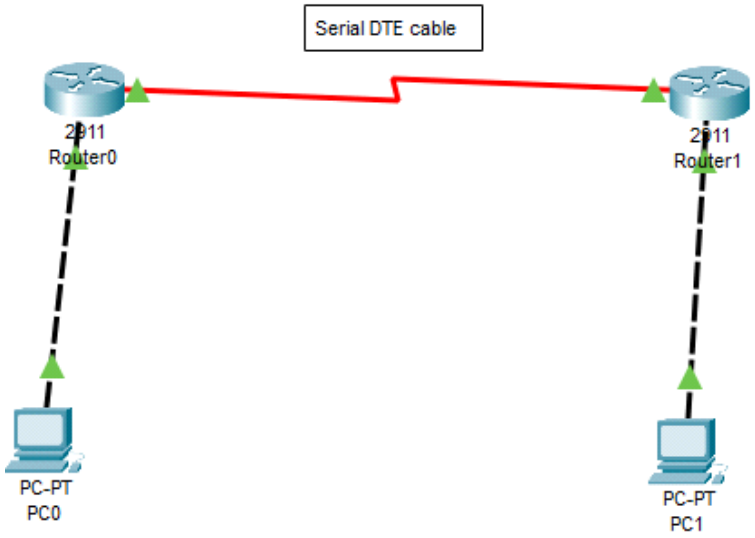
Pinging 200.100.100.0 with 32 bytes of data:

Reply from 2.1.1.1: bytes=32 time=7ms TTL=254
Reply from 2.1.1.1: bytes=32 time=5ms TTL=254
Reply from 2.1.1.1: bytes=32 time=1ms TTL=254
Reply from 2.1.1.1: bytes=32 time=10ms TTL=254

Ping statistics for 200.100.100.0:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 5ms

C:\>
```

- Final output image



Lab 8 : Decoy scan

i) IPS/IDS

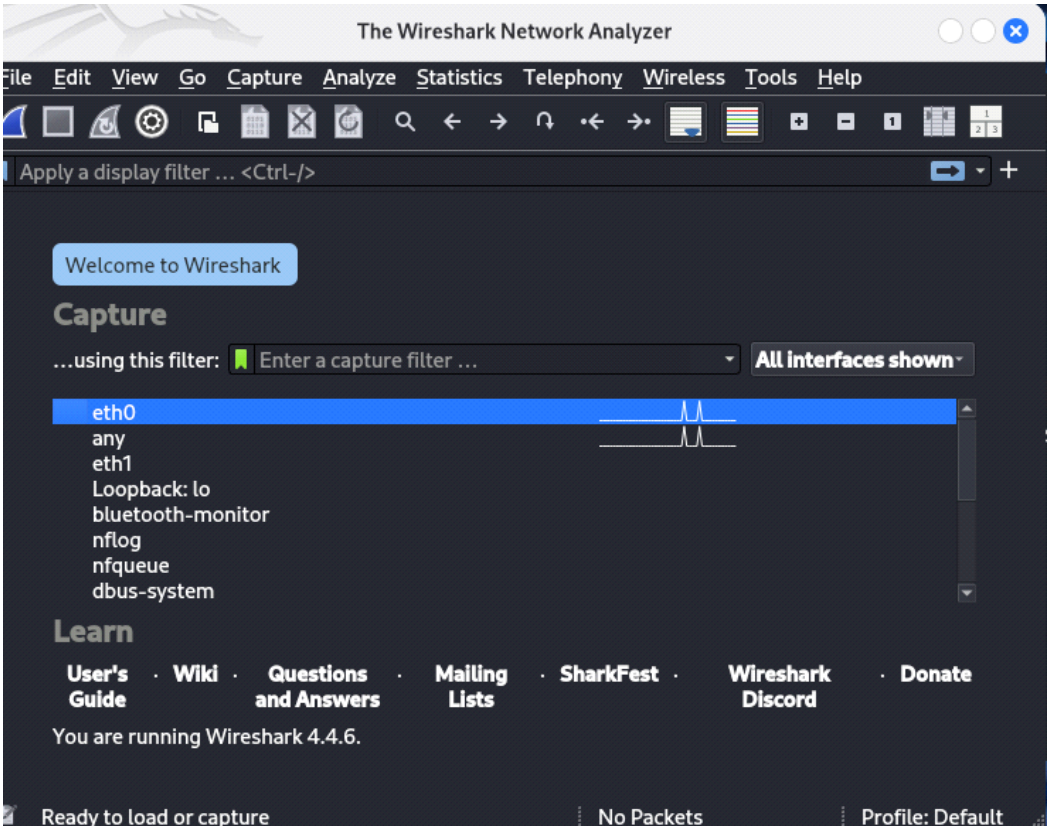
Allow well known ports to traffic

Nmap --source port

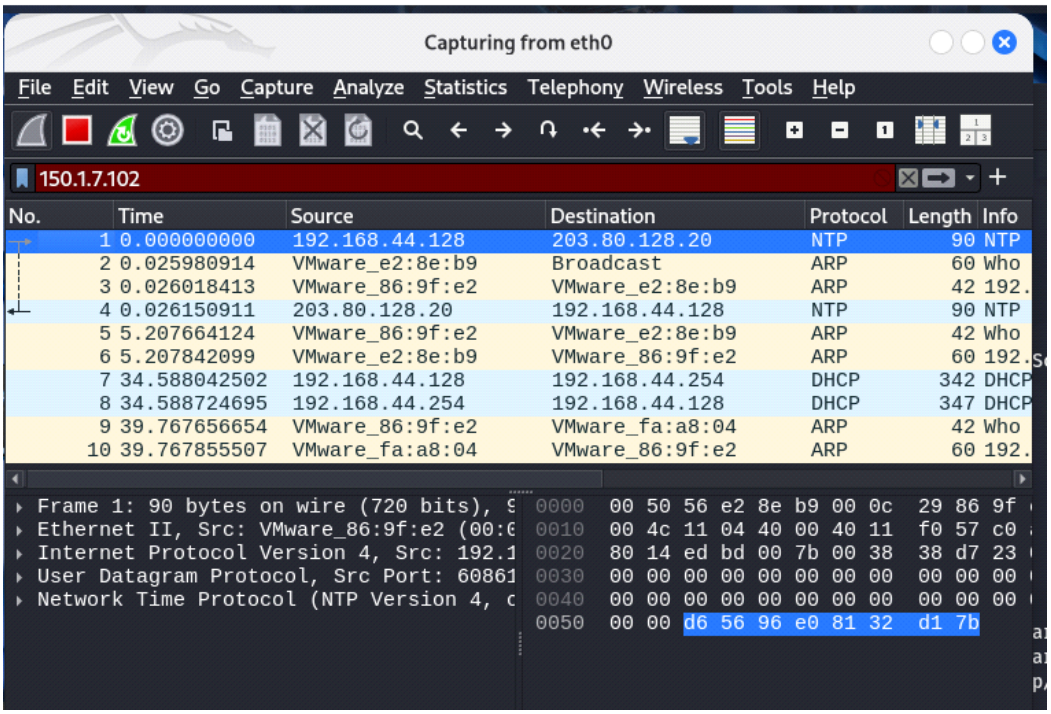
nmap -f 150.1.7.102 //f feagments

```
┌──(root㉿kali)-[/home/kali]
└─# sudo nmap -f 150.1.7.1.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 23:59 EDT
Failed to resolve "150.1.7.1.102".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 15.12 seconds
```

ii) Open wireshark with command sudo wireshark



Double click on eth0



```
┌──(root㉿kali)-[/home/kali]
└─# sudo wireshark
 ** (wireshark:4689) 09:12:01.901480 [Capture MESSAGE] -- Capture Start ...
 ** (wireshark:4689) 09:12:02.570817 [Capture MESSAGE] -- Capture started
 ** (wireshark:4689) 09:12:02.570869 [Capture MESSAGE] -- File: "/tmp/wireshark_
eth0XGHCF3.pcapng"
 ** (wireshark:4689) 09:17:55.277525 [Capture MESSAGE] -- Capture Stop ...
 ** (wireshark:4689) 09:17:55.473486 [Capture MESSAGE] -- Capture stopped.
 ** (wireshark:4689) 09:17:57.263440 [GUI WARNING] -- QThreadStorage: Thread 0x5
5724e5f8700 exited after QThreadStorage 4 destroyed
 ** (wireshark:4689) 09:17:57.263812 [GUI WARNING] -- QThreadStorage: Thread 0x5
5724e5f8700 exited after QThreadStorage 3 destroyed
 ** (wireshark:4689) 09:17:57.263861 [GUI WARNING] -- QThreadStorage: Thread 0x5
5724e5f8700 exited after QThreadStorage 2 destroyed

┌──(root㉿kali)-[/home/kali]
```

Enable wireshark

Ens 34

```
┌──(root㉿kali)-[/home/kali]
└─# nmap --mtu 16 150.1.7.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-02 09:19 EST
Nmap scan report for 150.1.7.102
Host is up (0.037s latency).
All 1000 scanned ports on 150.1.7.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 48.76 seconds

┌──(root㉿kali)-[/home/kali]
└─#
```

Decoy scan (specific decoy IPs)

```
┌──(kali㉿kali)-[~]
└─$ sudo su
[sudo] password for kali:
┌──(root㉿kali)-[/home/kali]
└─# nmap -D 150.1.7.102,150.1.7.120
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-03 21:16 EST
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.23 seconds

┌──(root㉿kali)-[/home/kali]
└─#
```

Random decoy count

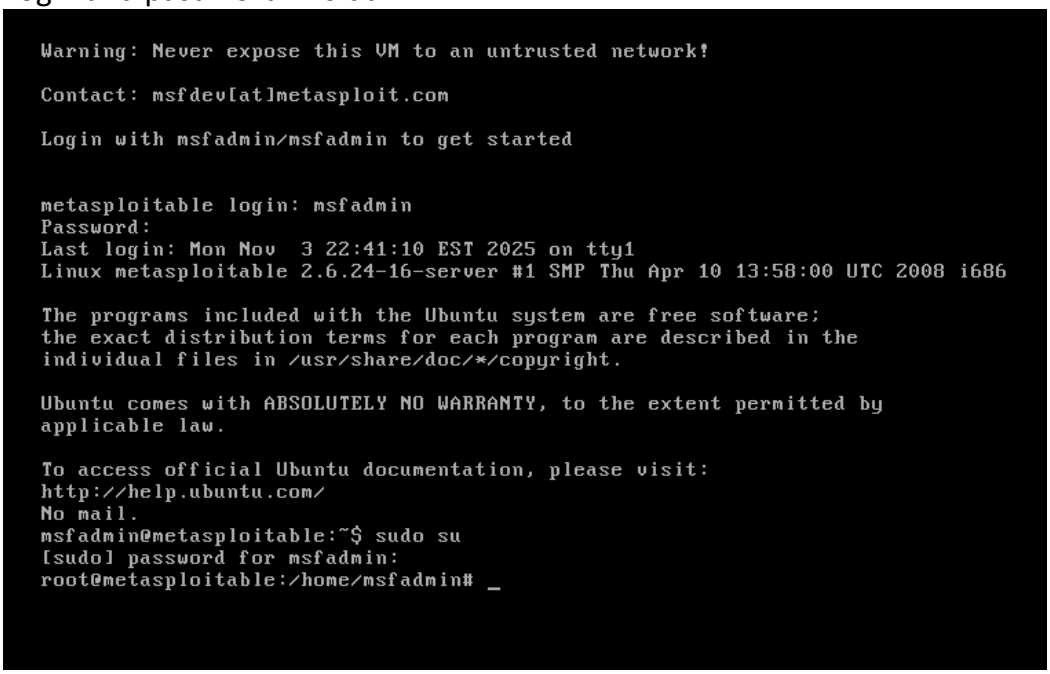
```
┌──(root㉿kali)-[/home/kali]
└─# nmap -D RND 150.1.7.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-03 21:21 EST
Nmap scan report for 150.1.7.102
Host is up (0.047s latency).
All 1000 scanned ports on 150.1.7.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 115.70 seconds

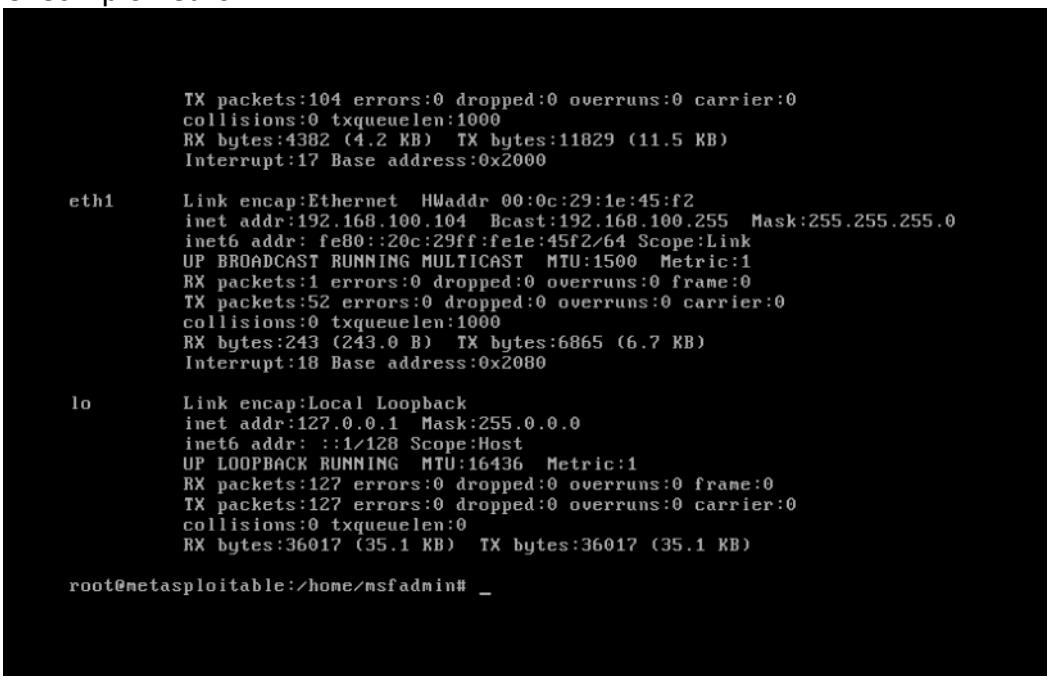
┌──(root㉿kali)-[/home/kali]
└─#
```

Lab 9 : Metasploitable

Login and pasdword: msfadmin



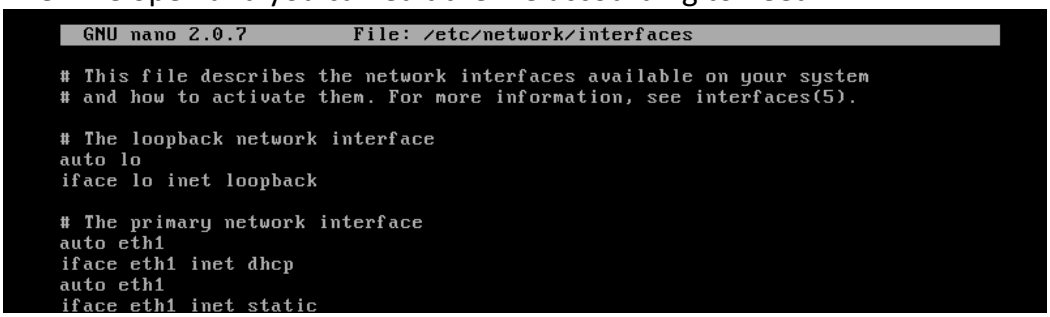
Check ip on eth0



Change ip on eth1

Write command sudo nano /etc/network/interfaces

Then file open and you can edit the file accourding to need.



```
  GNU nano 2.0.7        File: /etc/network/interfaces

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
auto eth1
iface eth1 inet dhcp
auto eth1
iface eth1 inet static
address 150.1.7.104
netmask 255.255.255.0
gateway 150.1.7.100




                        [ Wrote 19 lines ]

root@metasploitable:/home/msfadmin# _
```

Check the modification in file with the command
cat /etc/network/interfaces

```
root@metasploitable:/home/msfadmin# Reboot
bash: Reboot: command not found
root@metasploitable:/home/msfadmin# cat /etc/network/interfaces
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
auto eth1
iface eth1 inet dhcp
auto eth1
iface eth1 inet static
address 150.1.7.104
netmask 255.255.255.0
gateway 150.1.7.100




root@metasploitable:/home/msfadmin#
```

Assign ip to eth1 with command sudo ifconfig eth1 150.1.7.104 netmask 255.255.255.0
Then enter command ifconfig

```
          BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
          Interrupt:17 Base address:0x2000

eth1      Link encap:Ethernet  HWaddr 00:0c:29:1e:45:f2
          inet addr:150.1.7.104  Bcast:150.1.7.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe1e:45f2/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:6 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:468 (468.0 B)
          Interrupt:18 Base address:0x2080

lo        Link encap:Local Loopback
          LOOPBACK  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
```

```
lo        Link encap:Local Loopback
          LOOPBACK  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

root@metasploitable:/home/msfadmin# _
```

ping on kali

```
┌──(root㉿kali)-[/home/kali]
└─# ping 150.1.7.104
PING 150.1.7.104 (150.1.7.104) 56(84) bytes of data.
^C
--- 150.1.7.104 ping statistics ---
472 packets transmitted, 0 received, 100% packet loss, time 487575ms
```

Lab : Pyphisher

Sudo apt update

```
┌──(root㉿kali)-[/home/kali]
└─# sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.2 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [319 k
B]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [186 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [893
kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [1
1.3 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (d
eb) [28.4 kB]
Fetched 74.7 MB in 4min 57s (251 kB/s)
1808 packages can be upgraded. Run 'apt list --upgradable' to see them.

┌──(root㉿kali)-[/home/kali]
└─#
```

Sudo apt install curl

```
┌──(root㉿kali)-[/home]
└─# exit

┌──(kali㉿kali)-[~]
└─$ sudo apt install curl
[sudo] password for kali:
Upgrading:
  curl  libcurl3t64-gnutls  libcurl4t64

Summary:
  Upgrading: 3, Installing: 0, Removing: 0, Not Upgrading: 1805
  Download size: 0 B / 1,045 kB
  Space needed: 45.1 kB / 57.9 GB available

Continue? [Y/n] y
dpkg: warning: 'start-stop-daemon' not found in PATH or not executable
dpkg: error: 1 expected program not found in PATH or not executable
Note: root's PATH should usually contain /usr/local/sbin, /usr/sbin and /sbin
Error: Sub-process /var/tmp/dpkg-eatmydata-wrapper returned an error code (2)

┌──(kali㉿kali)-[~]
└─$ apt
```

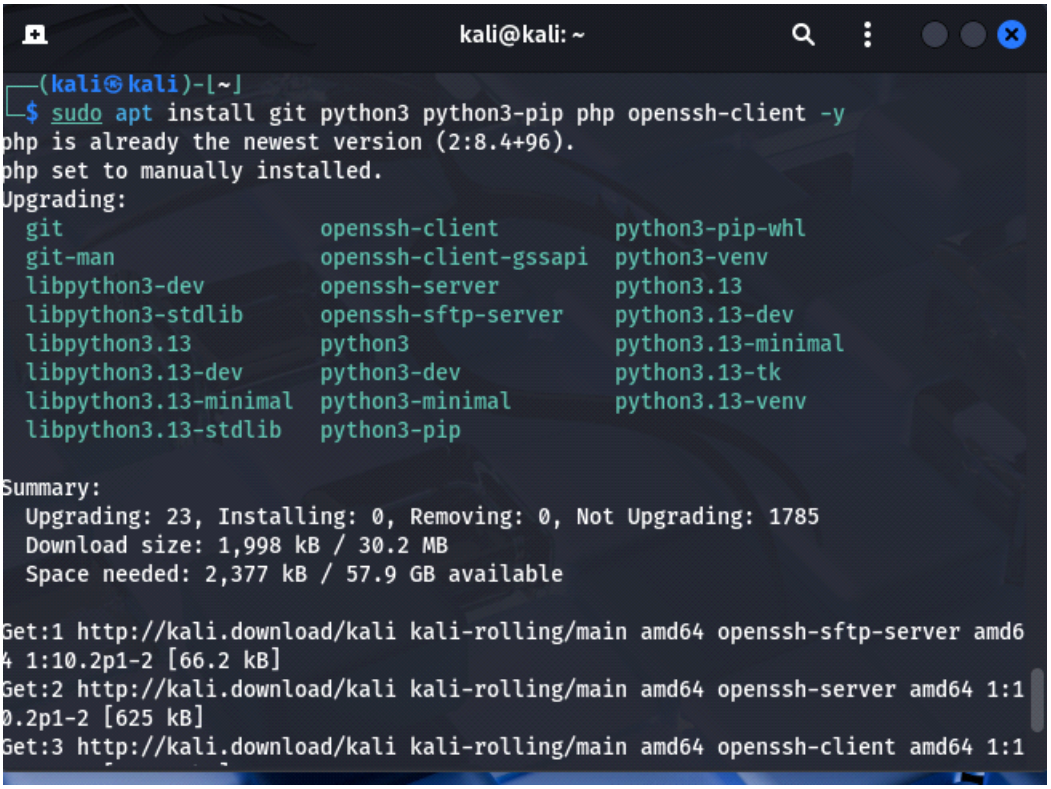
Sudo apt install openssl

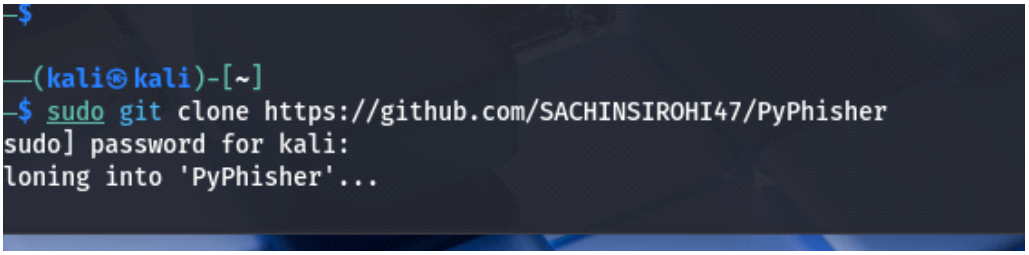
```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ sudo apt install openssl
Upgrading:
  libssl3t64  openssl  openssl-provider-legacy

Summary:
  Upgrading: 3, Installing: 0, Removing: 0, Not Upgrading: 1805
  Download size: 4,250 kB
  Freed space: 109 kB

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 openssl-provider-legacy
amd64 3.5.4-1 [308 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 libssl3t64 amd64 3.5.4-1
 [2,446 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 openssl amd64 3.5.4-1 [1
,496 kB]
Fetched 4,250 kB in 6s (701 kB/s)
dpkg: warning: 'start-stop-daemon' not found in PATH or not executable
dpkg: error: 1 expected program not found in PATH or not executable
Note: root's PATH should usually contain /usr/local/sbin, /usr/sbin and /sbin
Error: Sub-process /var/tmp/dpkg-eatmydata-wrapper returned an error code (2)

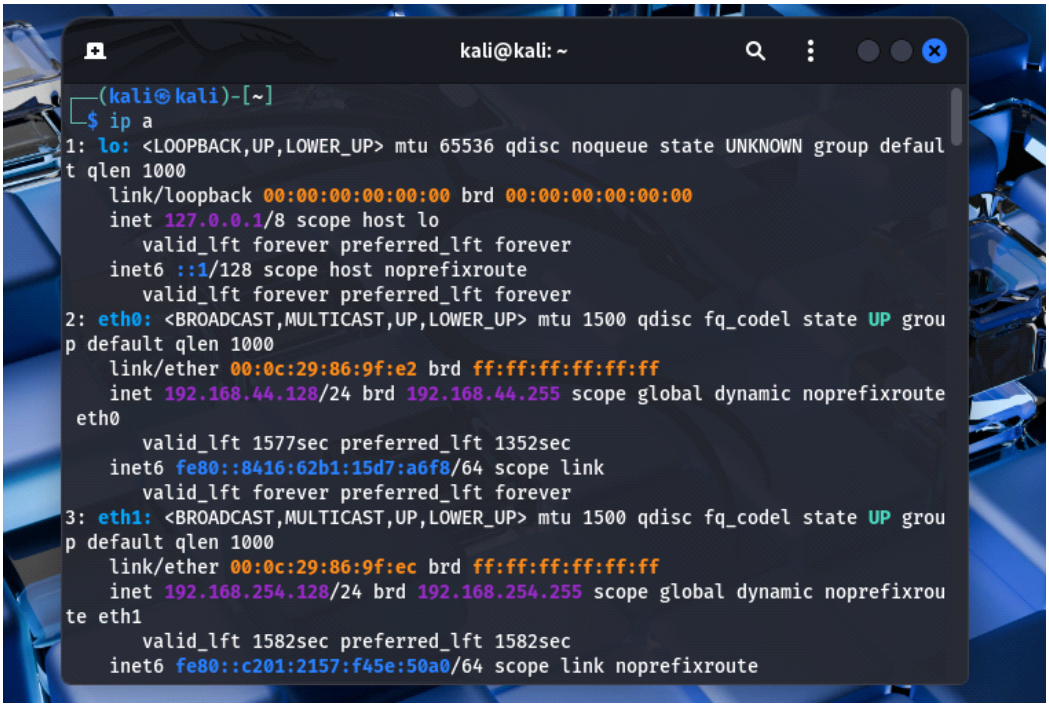
┌──(kali㉿kali)-[~]
└─
```

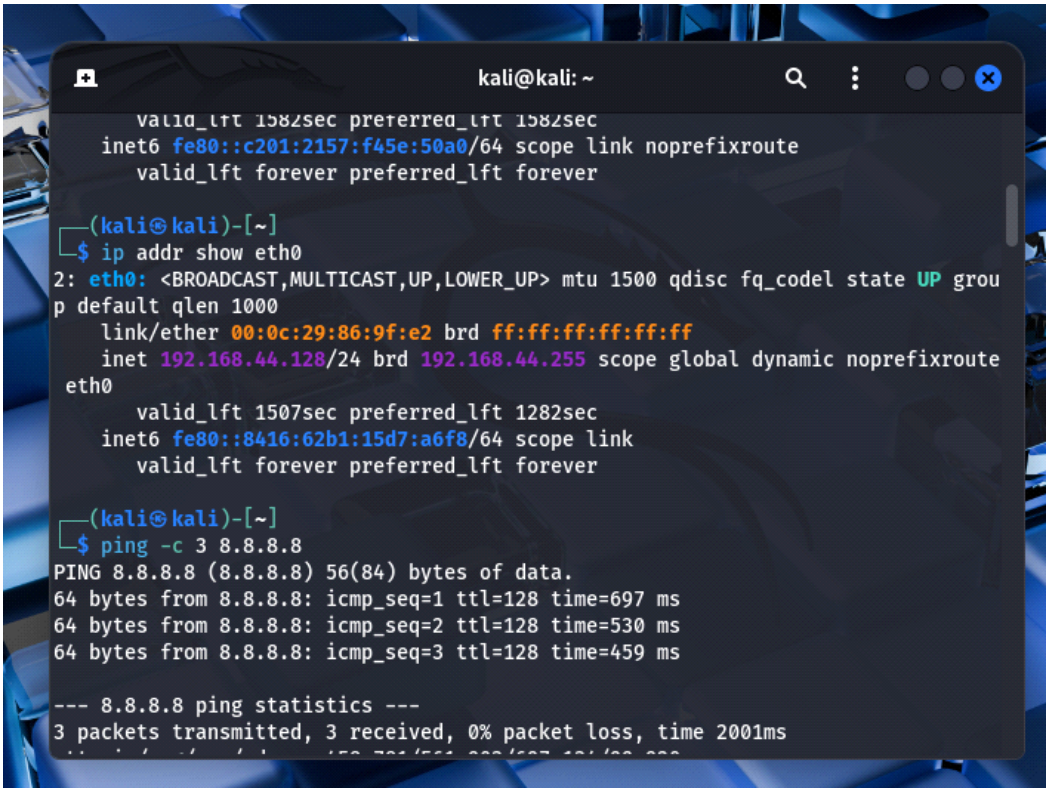


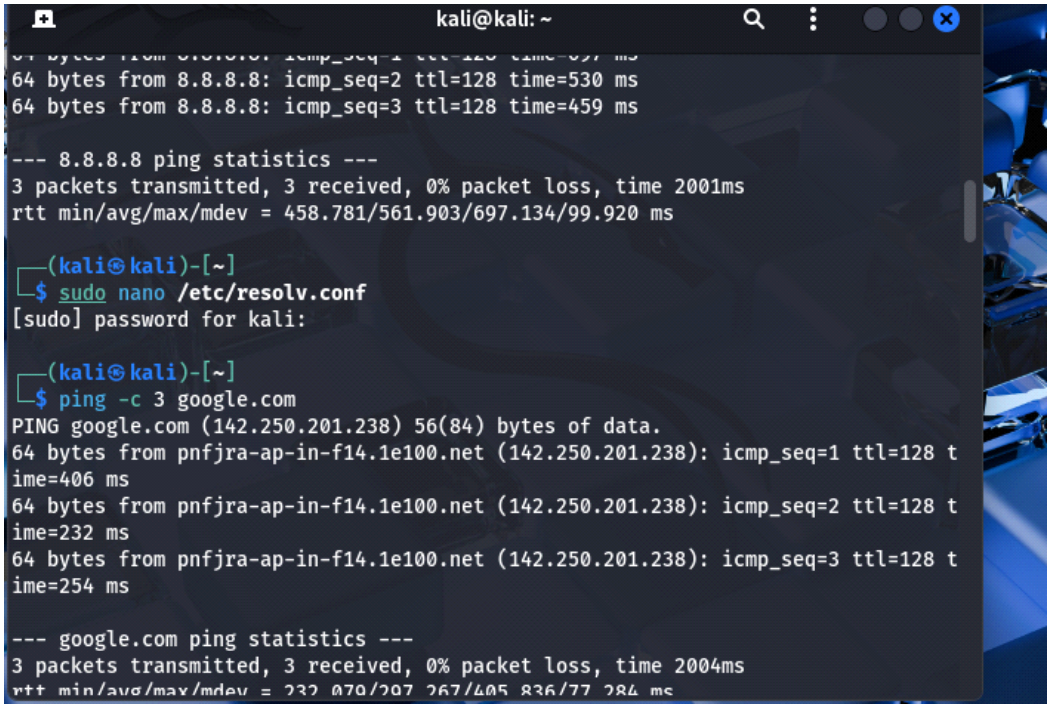


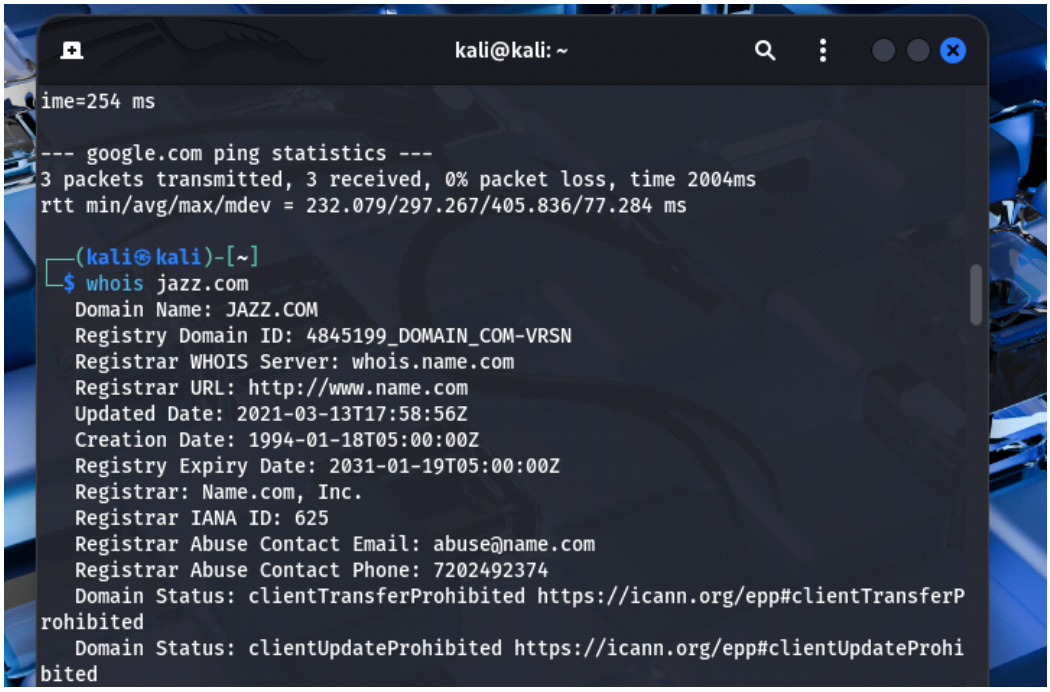
Information gathering :

1. Footprinting --> whois,nslookup,dig


kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:86:9f:e2 brd ff:ff:ff:ff:ff:ff
inet 192.168.44.128/24 brd 192.168.44.255 scope global dynamic noprefixroute eth0
valid_lft 1577sec preferred_lft 1352sec
inet6 fe80::8416:62b1:15d7:a6f8/64 scope link
valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:86:9f:ec brd ff:ff:ff:ff:ff:ff
inet 192.168.254.128/24 brd 192.168.254.255 scope global dynamic noprefixroute eth1
valid_lft 1582sec preferred_lft 1582sec
inet6 fe80::c201:2157:f45e:50a0/64 scope link noprefixroute


kali@kali: ~
valid_lft 1582sec preferred_lft 1582sec
inet6 fe80::c201:2157:f45e:50a0/64 scope link noprefixroute
valid_lft forever preferred_lft forever
┌──(kali㉿kali)-[~]
└─$ ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:86:9f:e2 brd ff:ff:ff:ff:ff:ff
inet 192.168.44.128/24 brd 192.168.44.255 scope global dynamic noprefixroute eth0
valid_lft 1507sec preferred_lft 1282sec
inet6 fe80::8416:62b1:15d7:a6f8/64 scope link
valid_lft forever preferred_lft forever
┌──(kali㉿kali)-[~]
└─$ ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=697 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=530 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=459 ms
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2001ms


kali@kali: ~
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=530 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=459 ms
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2001ms
rtt min/avg/max/mdev = 458.781/561.903/697.134/99.920 ms
┌──(kali㉿kali)-[~]
└─$ sudo nano /etc/resolv.conf
[sudo] password for kali:
┌──(kali㉿kali)-[~]
└─$ ping -c 3 google.com
PING google.com (142.250.201.238) 56(84) bytes of data.
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=1 ttl=128 time=406 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=2 ttl=128 time=232 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=3 ttl=128 time=254 ms
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms


kali@kali: ~
ime=254 ms
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 232.079/297.267/405.836/77.284 ms
┌──(kali㉿kali)-[~]
└─$ whois jazz.com
Domain Name: JAZZ.COM
Registry Domain ID: 4845199_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.name.com
Registrar URL: http://www.name.com
Updated Date: 2021-03-13T17:58:56Z
Creation Date: 1994-01-18T05:00:00Z
Registry Expiry Date: 2031-01-19T05:00:00Z
Registrar: Name.com, Inc.
Registrar IANA ID: 625
Registrar Abuse Contact Email: abuse@name.com
Registrar Abuse Contact Phone: 7202492374
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited

```
┌──(kali㉿kali)-[~]
└─$

┌──(kali㉿kali)-[~]
└─$ nslookup covrit.com
Server:         192.168.44.2
Address:        192.168.44.2#53

Non-authoritative answer:
Name:   covrit.com
Address: 35.215.77.113


┌──(kali㉿kali)-[~]
└─$
```

```
└─$ dig zong.com

; <<>> DiG 9.20.9-1-Debian <<>> zong.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 21535
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;zong.com.                      IN      A

;; ANSWER SECTION:
zong.com.               5       IN      A       99.83.176.46

;; Query time: 592 msec
;; SERVER: 192.168.44.2#53(192.168.44.2) (UDP)
;; WHEN: Fri Nov 07 00:15:26 EST 2025
;; MSG SIZE  rcvd: 53


┌──(kali㉿kali)-[~]
└─$
```

2.scanning
Host discovery

```
┌──(kali㉿kali)-[~]
└─$

┌──(kali㉿kali)-[~]
└─$ nmap -sn 192.168.44.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-07 00:22 EST
Nmap scan report for 192.168.44.1
Host is up (0.0013s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.44.2
Host is up (0.00070s latency).
MAC Address: 00:50:56:E2:8E:B9 (VMware)
Nmap scan report for 192.168.44.254
Host is up (0.00056s latency).
MAC Address: 00:50:56:E2:40:AA (VMware)
Nmap scan report for 192.168.44.128
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.81 seconds

┌──(kali㉿kali)-[~]
└─$
```

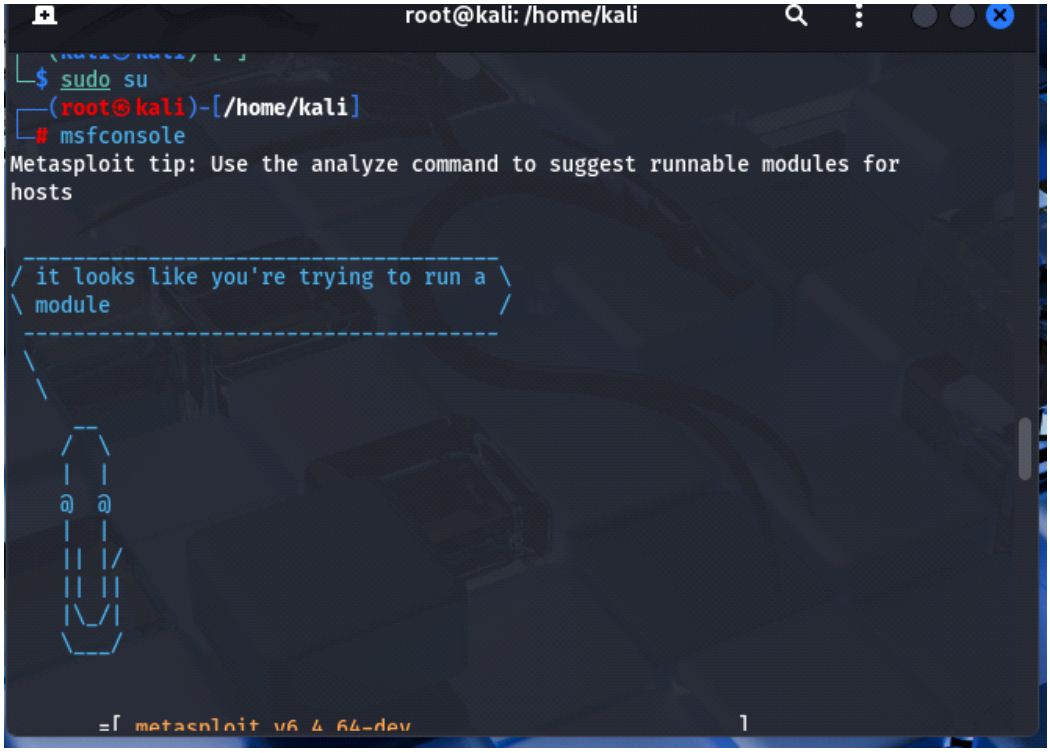
Ports scanning

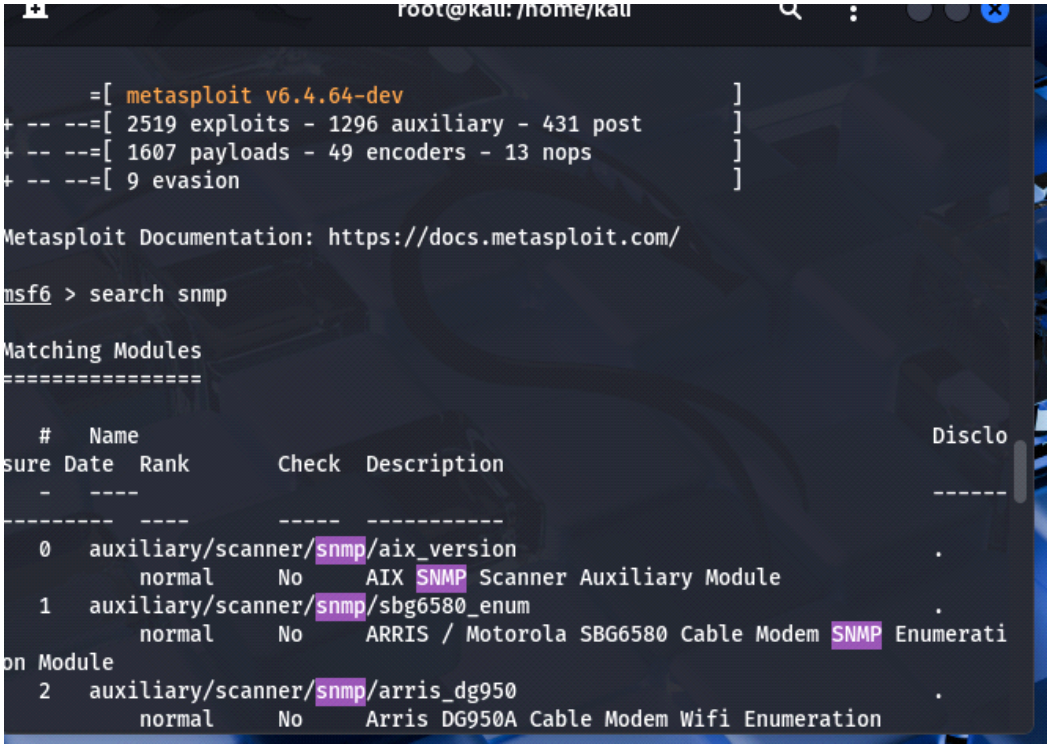
```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -T4 192.168.44.128
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-07 00:25 EST
Nmap scan report for 192.168.44.128
Host is up (0.000012s latency).
All 1000 scanned ports on 192.168.44.128 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

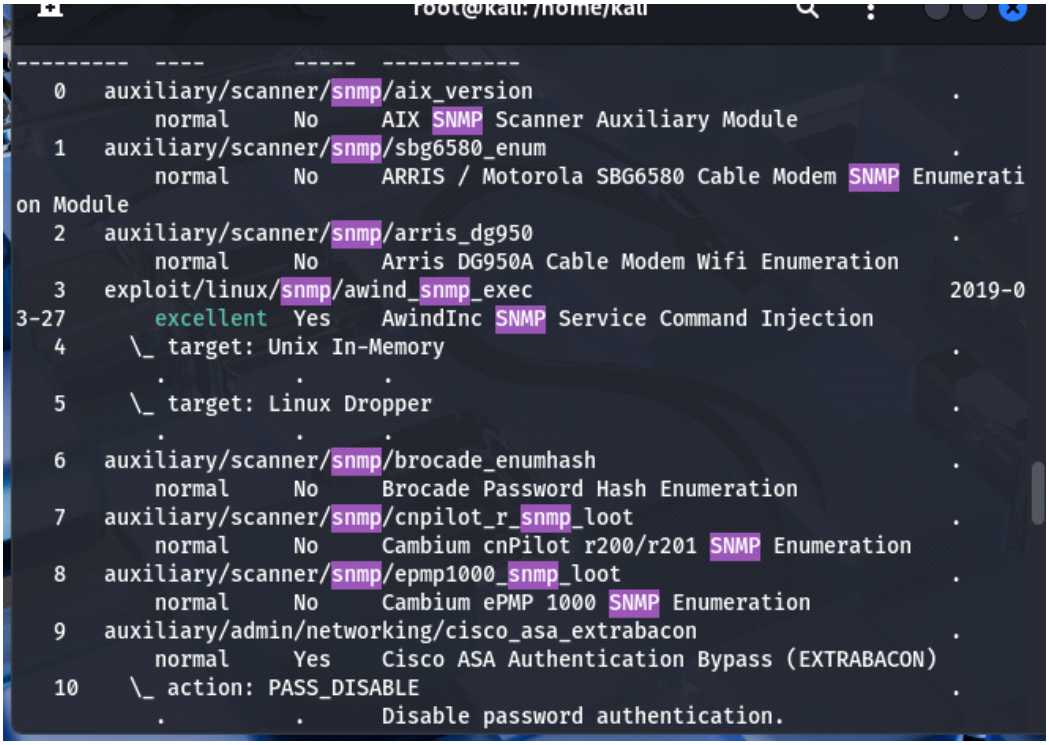
Nmap done: 1 IP address (1 host up) scanned in 2.50 seconds

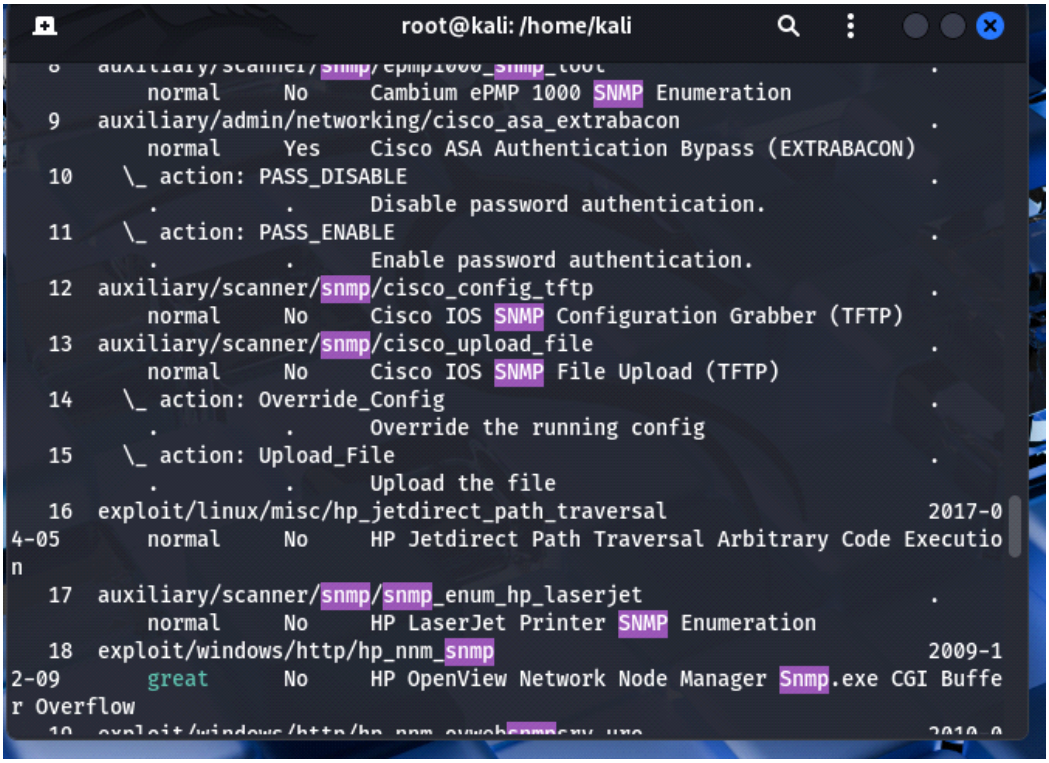
┌──(kali㉿kali)-[~]
└─$
```

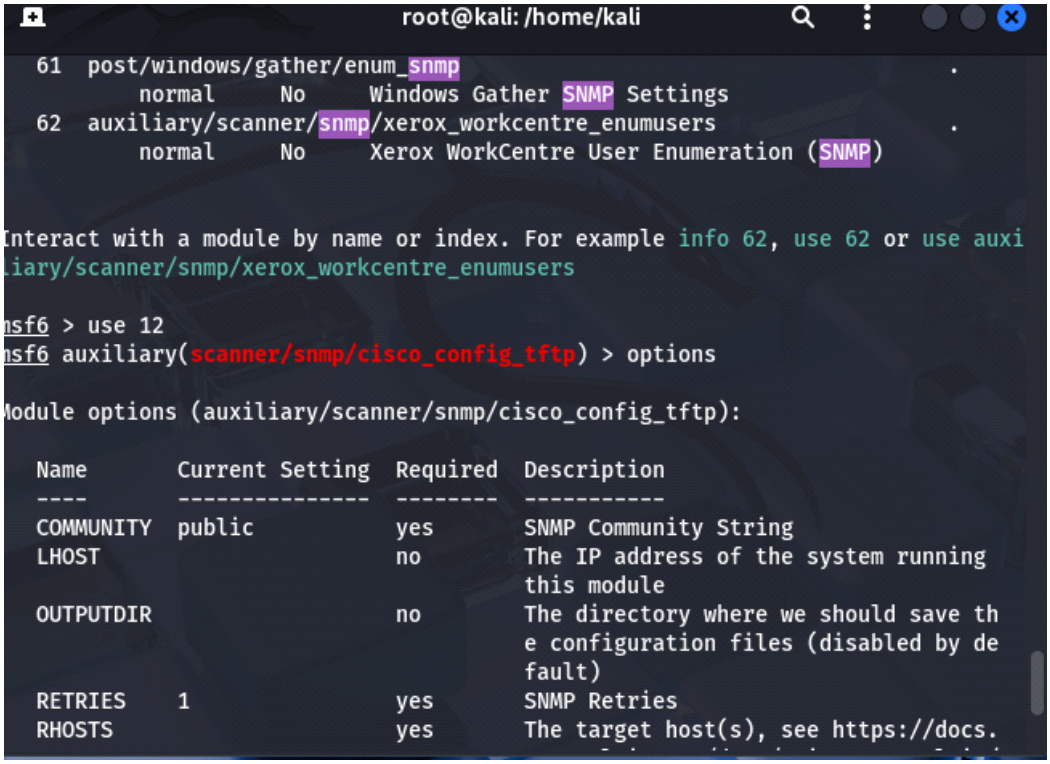
SNMP enumaration

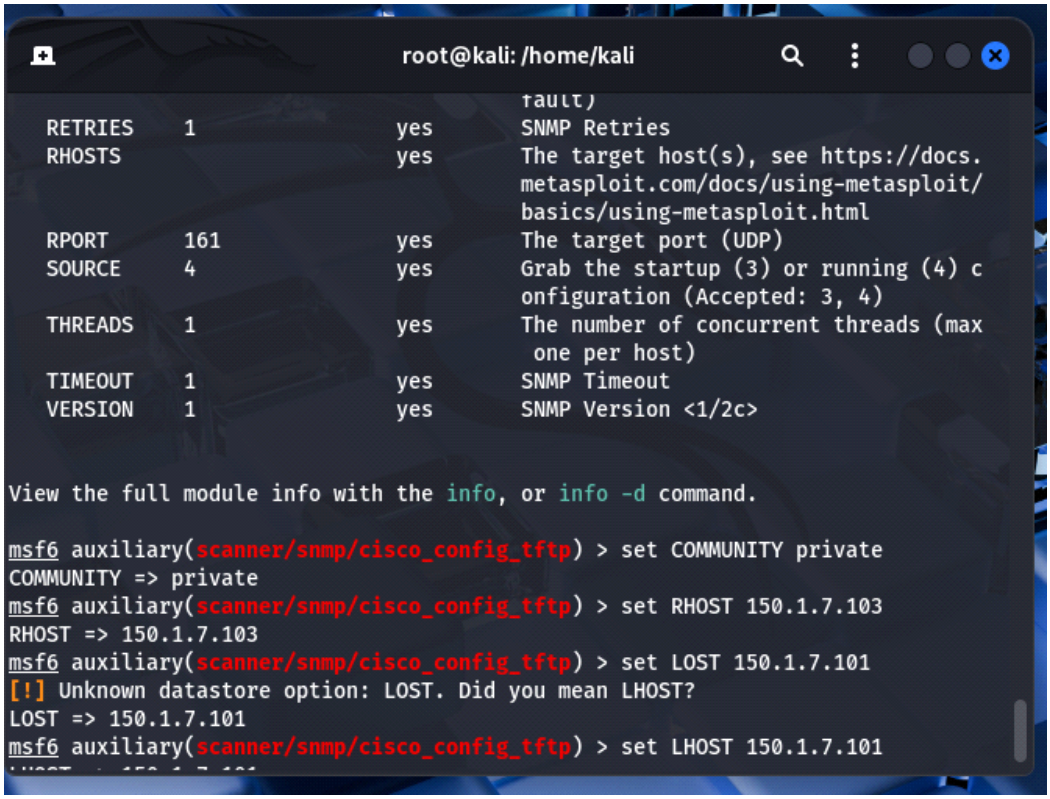

root@kali: /home/kali
$ sudo su
(root㉿kali)-[/home/kali]
msfconsole
Metasploit tip: Use the analyze command to suggest runnable modules for hosts
/ it looks like you're trying to run a \
\ module /
=[metasploit v6.4.64-dev]

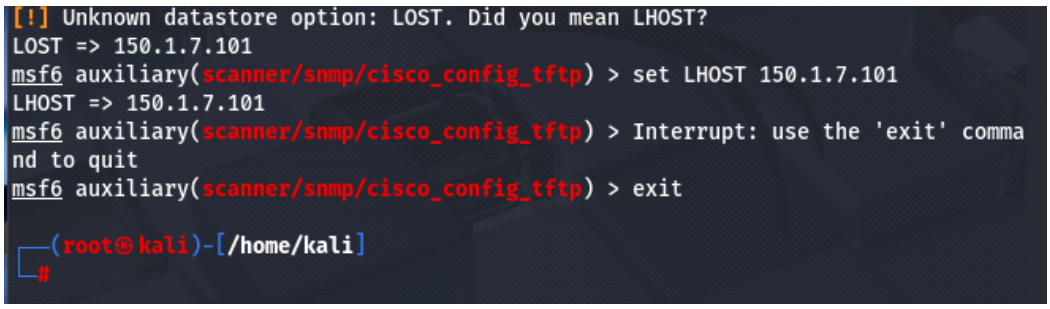

root@kali: /home/kali
=[metasploit v6.4.64-dev]
+ -- --=[2519 exploits - 1296 auxiliary - 431 post]
+ -- --=[1607 payloads - 49 encoders - 13 nops]
+ -- --=[9 evasion]
Metasploit Documentation: https://docs.metasploit.com/
msf6 > search snmp
Matching Modules
Name Disclosure Date Rank Check Description
0 auxiliary/scanner/snmp/aix_version . normal No AIX SNMP Scanner Auxiliary Module
1 auxiliary/scanner/snmp/sbg6580_enum . normal No ARRIS / Motorola SBG6580 Cable Modem SNMP Enumeration Module
2 auxiliary/scanner/snmp/arris_dg950 . normal No Arris DG950A Cable Modem Wifi Enumeration


root@kali: /home/kali
--------- ---- ----- -----------
0 auxiliary/scanner/snmp/aix_version .
normal No AIX SNMP Scanner Auxiliary Module
1 auxiliary/scanner/snmp/sbg6580_enum .
normal No ARRIS / Motorola SBG6580 Cable Modem SNMP Enumerati
on Module
2 auxiliary/scanner/snmp/arris_dg950 .
normal No Arris DG950A Cable Modem Wifi Enumeration
3 exploit/linux/snmp/awind_snmp_exec 2019-0
3-27 excellent Yes AwindInc SNMP Service Command Injection
4 _ target: Unix In-Memory .
. . .
5 _ target: Linux Dropper .
. . .
6 auxiliary/scanner/snmp/brocade_enumhash .
normal No Brocade Password Hash Enumeration
7 auxiliary/scanner/snmp/cnpilot_r_snmp_loot .
normal No Cambium cnPilot r200/r201 SNMP Enumeration
8 auxiliary/scanner/snmp/epmp1000_snmp_loot .
normal No Cambium ePMP 1000 SNMP Enumeration
9 auxiliary/admin/networking/cisco_asa_extrabacon .
normal Yes Cisco ASA Authentication Bypass (EXTRABACON)
10 _ action: PASS_DISABLE .
. . Disable password authentication.

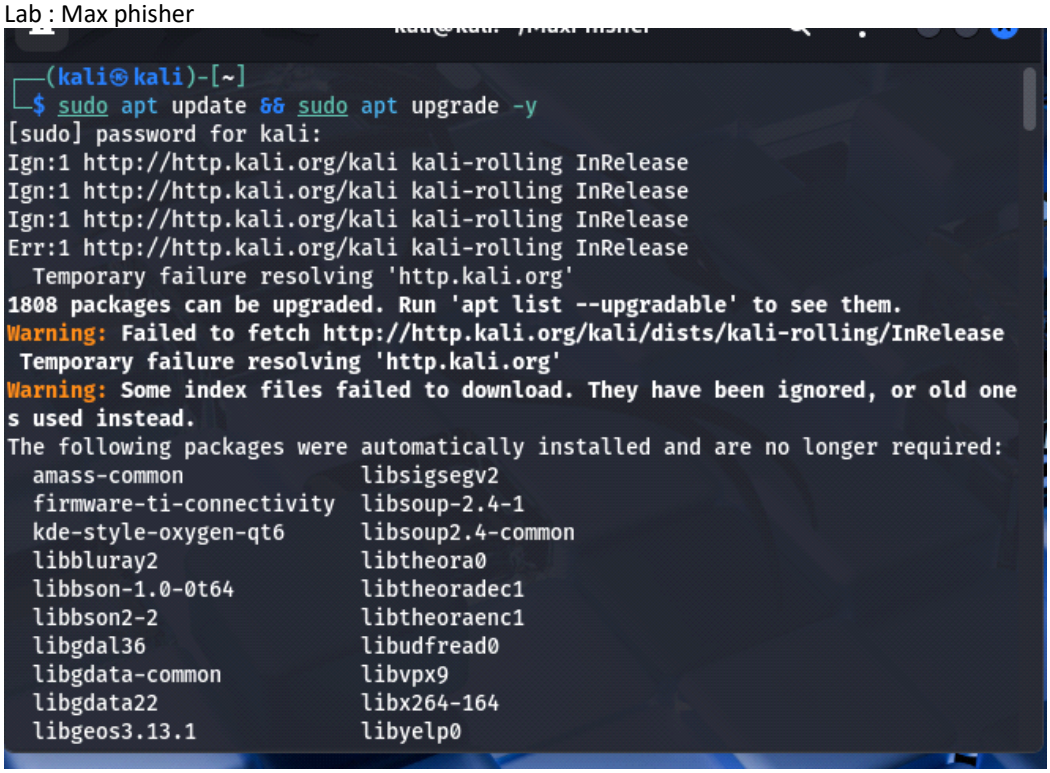

root@kali: /home/kali
normal No Cambium ePMP 1000 SNMP Enumeration
9 auxiliary/admin/networking/cisco_asa_extrabacon .
normal Yes Cisco ASA Authentication Bypass (EXTRABACON)
10 _ action: PASS_DISABLE .
. . Disable password authentication.
11 _ action: PASS_ENABLE .
. . Enable password authentication.
12 auxiliary/scanner/snmp/cisco_config_tftp .
normal No Cisco IOS SNMP Configuration Grabber (TFTP)
13 auxiliary/scanner/snmp/cisco_upload_file .
normal No Cisco IOS SNMP File Upload (TFTP)
14 _ action: Override_Config .
. . Override the running config
15 _ action: Upload_File .
. . Upload the file
16 exploit/linux/misc/hp_jetdirect_path_traversal 2017-0
4-05 normal No HP Jetdirect Path Traversal Arbitrary Code Executio
n
17 auxiliary/scanner/snmp/snmp_enum_hp_laserjet .
normal No HP LaserJet Printer SNMP Enumeration
18 exploit/windows/http/hp_nnm_snmp 2009-1
2-09 great No HP OpenView Network Node Manager Snmp.exe CGI Buffe
r Overflow

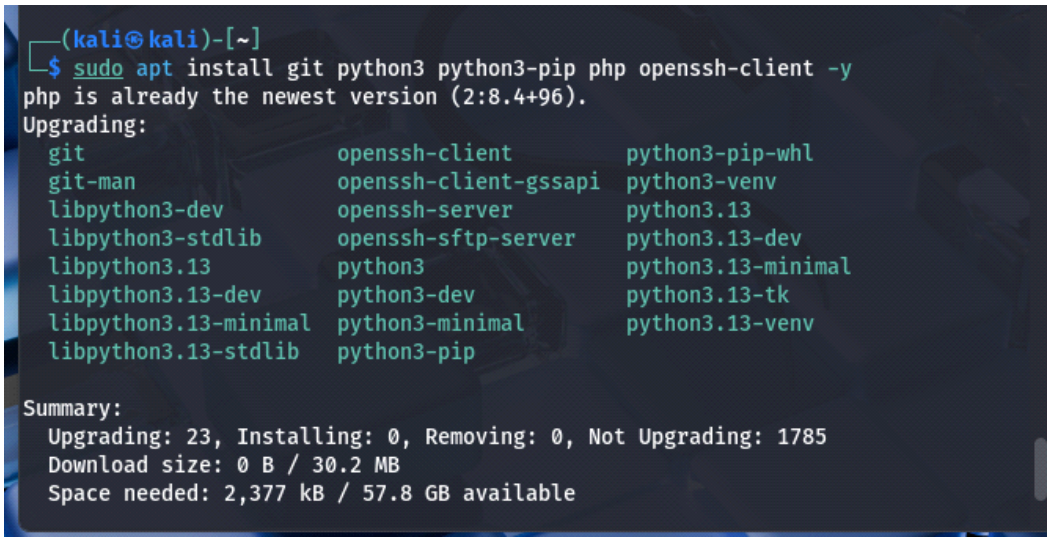

root@kali: /home/kali
61 post/windows/gather/enum_snmp
normal No Windows Gather SNMP Settings
62 auxiliary/scanner/snmp/xerox_workcentre_enumusers
normal No Xerox WorkCentre User Enumeration (SNMP)
Interact with a module by name or index. For example info 62, use 62 or use auxiliary/scanner/snmp/xerox_workcentre_enumusers
msf6 > use 12
msf6 auxiliary(scanner/snmp/cisco_config_tftp) > options
Module options (auxiliary/scanner/snmp/cisco_config_tftp):
Name Current Setting Required Description
COMMUNITY public yes SNMP Community String
LHOST no The IP address of the system running this module
OUTPUTDIR no The directory where we should save the configuration files (disabled by default)
RETRIES 1 yes SNMP Retries
RHOSTS yes The target host(s), see https://docs.

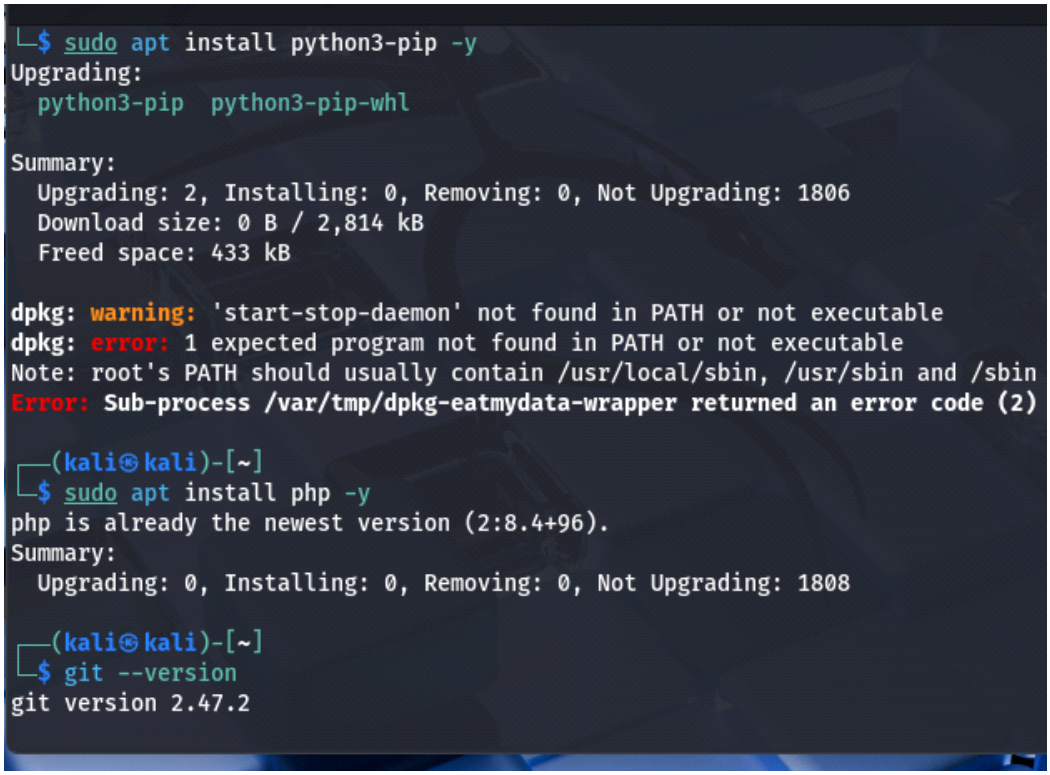

root@kali: /home/kali
RETRIES 1 yes SNMP Retries
RHOSTS yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 161 yes The target port (UDP)
SOURCE 4 yes Grab the startup (3) or running (4) configuration (Accepted: 3, 4)
THREADS 1 yes The number of concurrent threads (max one per host)
TIMEOUT 1 yes SNMP Timeout
VERSION 1 yes SNMP Version <1/2c>
View the full module info with the info, or info -d command.
msf6 auxiliary(scanner/snmp/cisco_config_tftp) > set COMMUNITY private
COMMUNITY => private
msf6 auxiliary(scanner/snmp/cisco_config_tftp) > set RHOST 150.1.7.103
RHOST => 150.1.7.103
msf6 auxiliary(scanner/snmp/cisco_config_tftp) > set LOST 150.1.7.101
[!] Unknown datastore option: LOST. Did you mean LHOST?
LOST => 150.1.7.101
msf6 auxiliary(scanner/snmp/cisco_config_tftp) > set LHOST 150.1.7.101

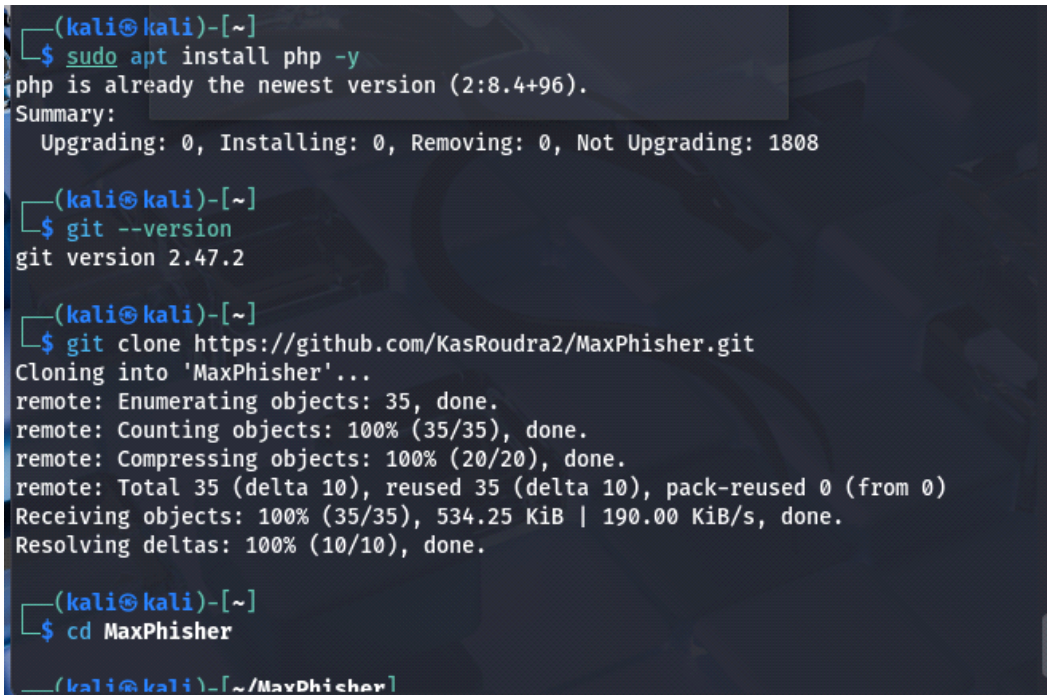

[!] Unknown datastore option: LOST. Did you mean LHOST?
LOST => 150.1.7.101
msf6 auxiliary(scanner/snmp/cisco_config_tftp) > set LHOST 150.1.7.101
LHOST => 150.1.7.101
msf6 auxiliary(scanner/snmp/cisco_config_tftp) > Interrupt: use the 'exit' command to quit
msf6 auxiliary(scanner/snmp/cisco_config_tftp) > exit
(root㉿kali)-[/home/kali]
#

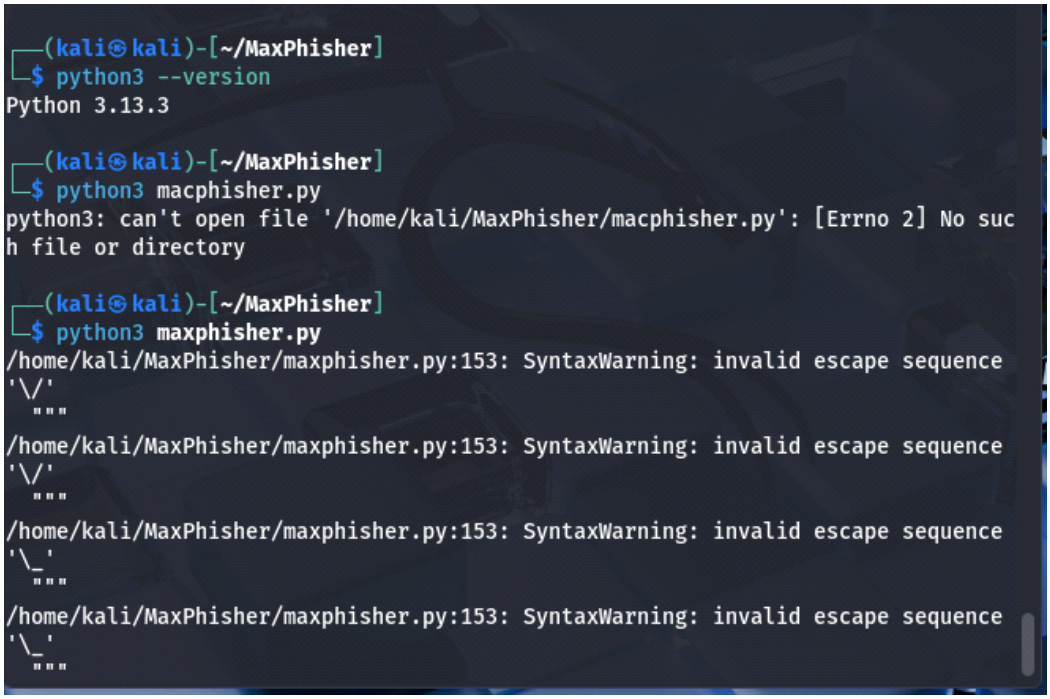
Lab : Max phisher

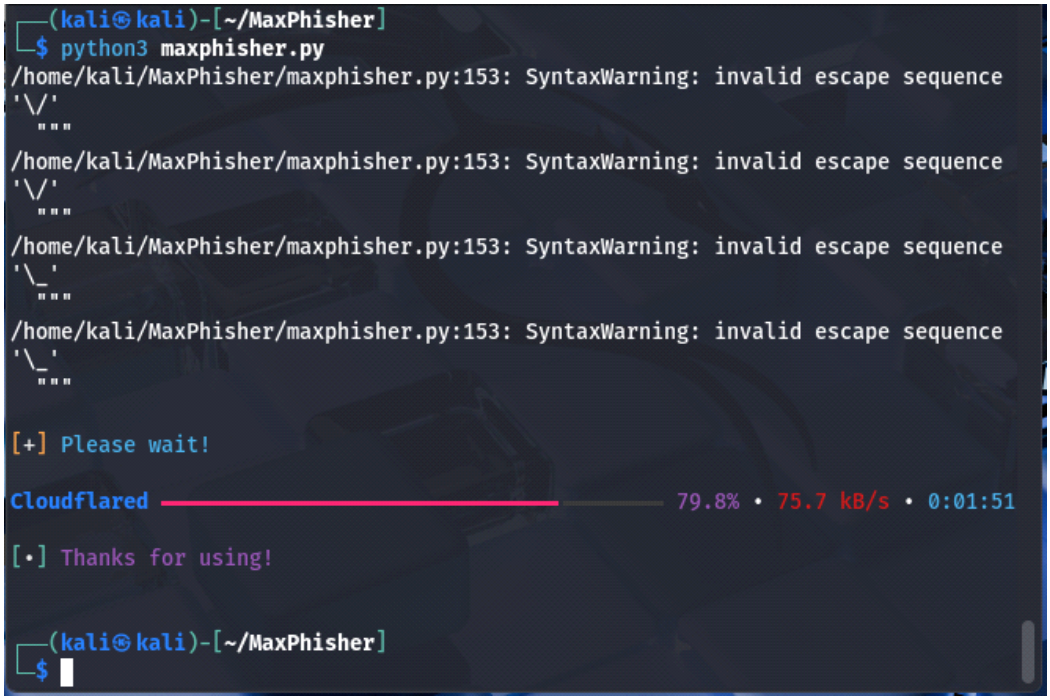




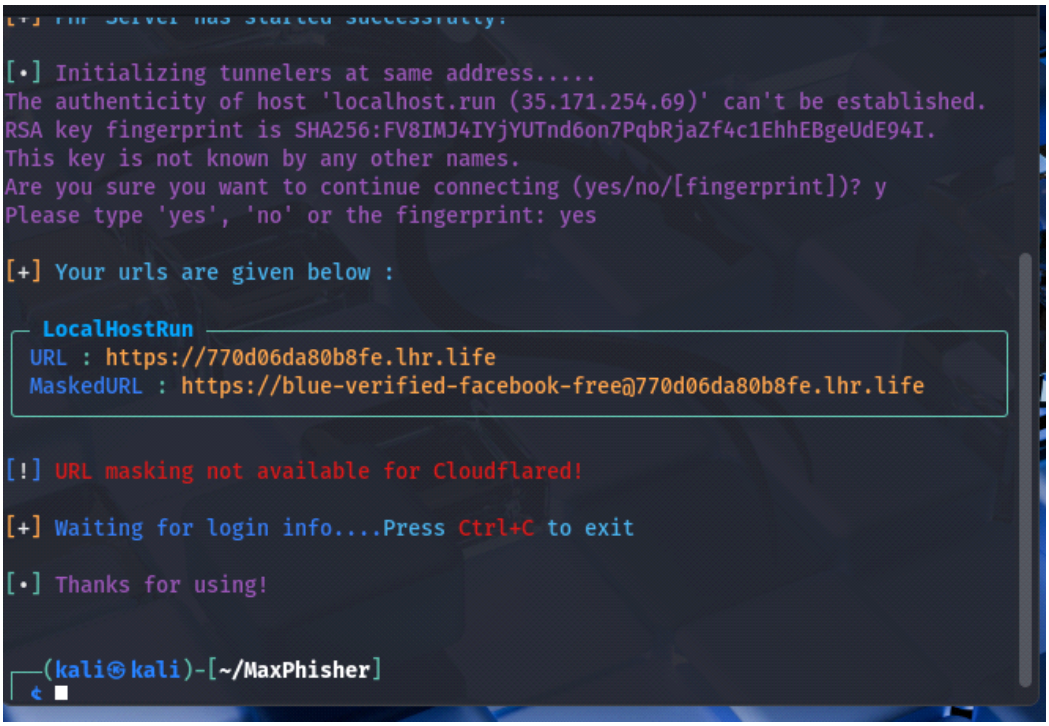

└─$ sudo apt install python3-pip -y
Upgrading:
python3-pip python3-pip-whl
Summary:
Upgrading: 2, Installing: 0, Removing: 0, Not Upgrading: 1806
Download size: 0 B / 2,814 kB
Freed space: 433 kB
dpkg: warning: 'start-stop-daemon' not found in PATH or not executable
dpkg: error: 1 expected program not found in PATH or not executable
Note: root's PATH should usually contain /usr/local/sbin, /usr/sbin and /sbin
Error: Sub-process /var/tmp/dpkg-eatmydata-wrapper returned an error code (2)
┌──(kali㉿kali)-[~]
└─$ sudo apt install php -y
php is already the newest version (2:8.4+96).
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1808
┌──(kali㉿kali)-[~]
└─$ git --version
git version 2.47.2


┌──(kali㉿kali)-[~]
└─$ sudo apt install php -y
php is already the newest version (2:8.4+96).
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1808
┌──(kali㉿kali)-[~]
└─$ git --version
git version 2.47.2
┌──(kali㉿kali)-[~]
└─$ git clone https://github.com/KasRoudra2/MaxPhisher.git
Cloning into 'MaxPhisher'...
remote: Enumerating objects: 35, done.
remote: Counting objects: 100% (35/35), done.
remote: Compressing objects: 100% (20/20), done.
remote: Total 35 (delta 10), reused 35 (delta 10), pack-reused 0 (from 0)
Receiving objects: 100% (35/35), 534.25 KiB | 190.00 KiB/s, done.
Resolving deltas: 100% (10/10), done.
┌──(kali㉿kali)-[~]
└─$ cd MaxPhisher


┌──(kali㉿kali)-[~/MaxPhisher]
└─$ python3 --version
Python 3.13.3
┌──(kali㉿kali)-[~/MaxPhisher]
└─$ python3 macphisher.py
python3: can't open file '/home/kali/MaxPhisher/macphisher.py': [Errno 2] No suc
h file or directory
┌──(kali㉿kali)-[~/MaxPhisher]
└─$ python3 maxphisher.py
/home/kali/MaxPhisher/maxphisher.py:153: SyntaxWarning: invalid escape sequence
'\/'
"""
/home/kali/MaxPhisher/maxphisher.py:153: SyntaxWarning: invalid escape sequence
'\/'
"""
/home/kali/MaxPhisher/maxphisher.py:153: SyntaxWarning: invalid escape sequence
'_'
"""
/home/kali/MaxPhisher/maxphisher.py:153: SyntaxWarning: invalid escape sequence
'_'
"""


┌──(kali㉿kali)-[~/MaxPhisher]
└─$ python3 maxphisher.py
/home/kali/MaxPhisher/maxphisher.py:153: SyntaxWarning: invalid escape sequence
'\/'
"""
/home/kali/MaxPhisher/maxphisher.py:153: SyntaxWarning: invalid escape sequence
'\/'
"""
/home/kali/MaxPhisher/maxphisher.py:153: SyntaxWarning: invalid escape sequence
'_'
"""
/home/kali/MaxPhisher/maxphisher.py:153: SyntaxWarning: invalid escape sequence
'_'
"""
[+] Please wait!
Cloudflared 79.8% • 75.7 kB/s • 0:01:51
[•] Thanks for using!
┌──(kali㉿kali)-[~/MaxPhisher]
└─$





Lab : share folder

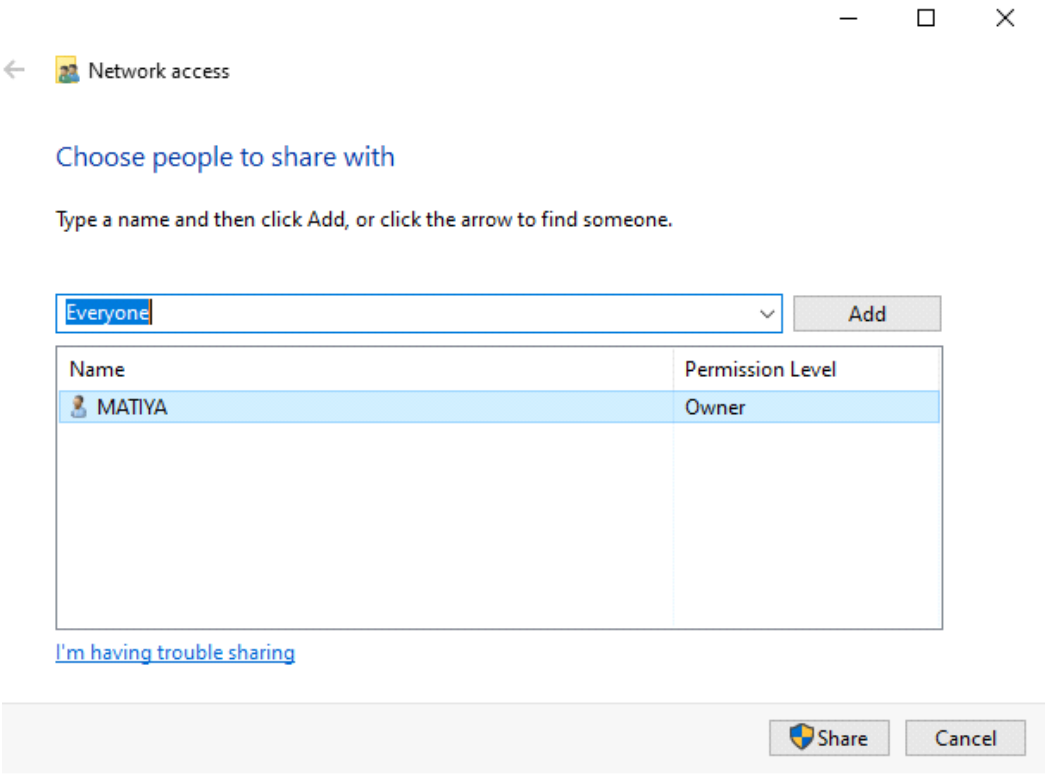
Purpose of share folder is that If we are on the same network, we can share files from one PC to another without using a USB, data cable, or any physical connection.

Create a folder

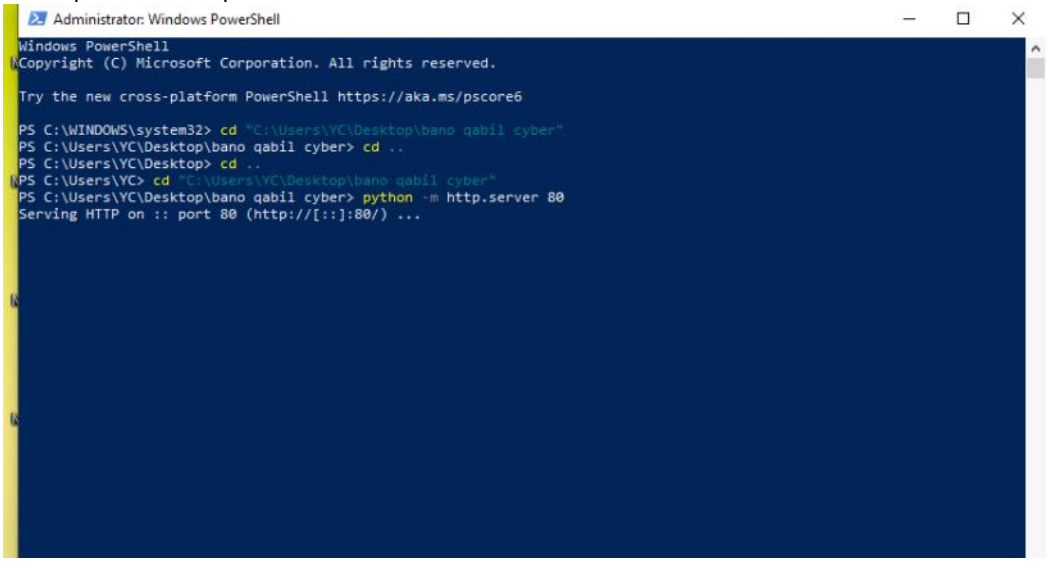
Right click on the folder

Go to give access

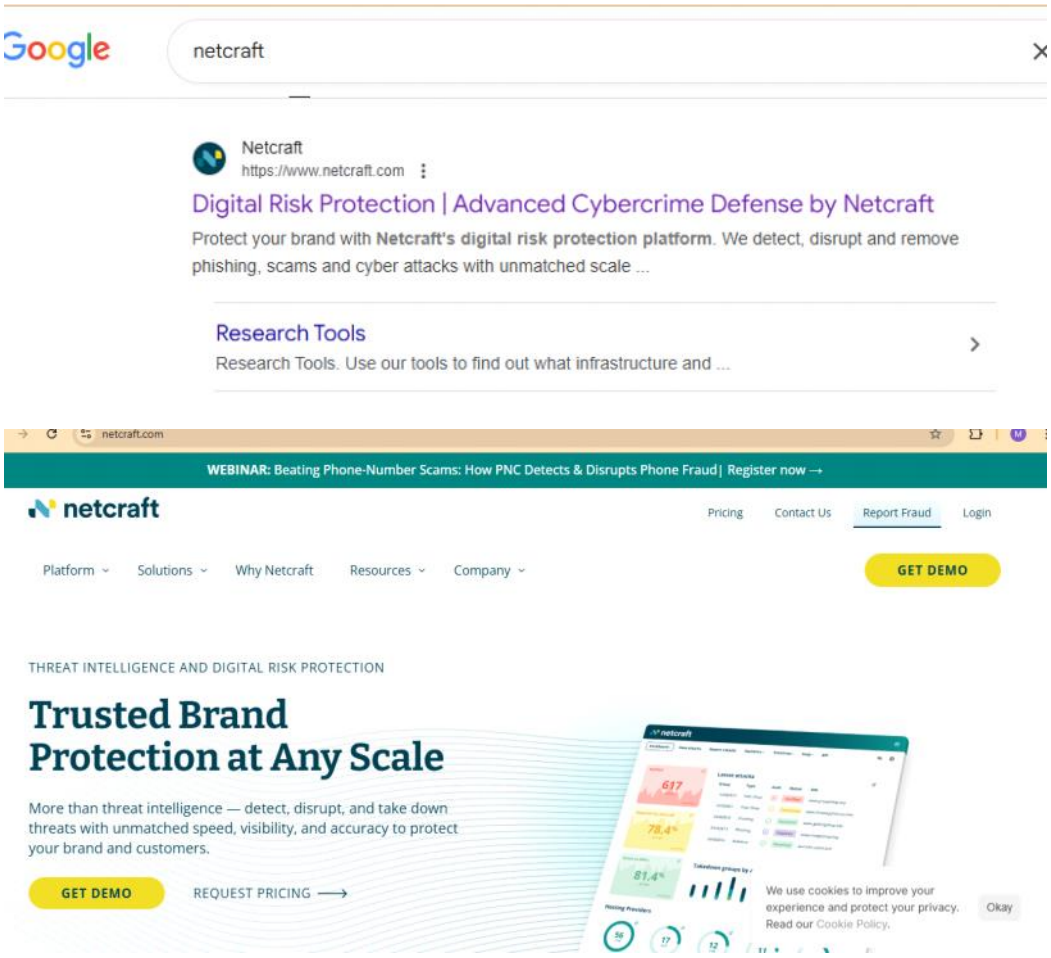
Choose specific people



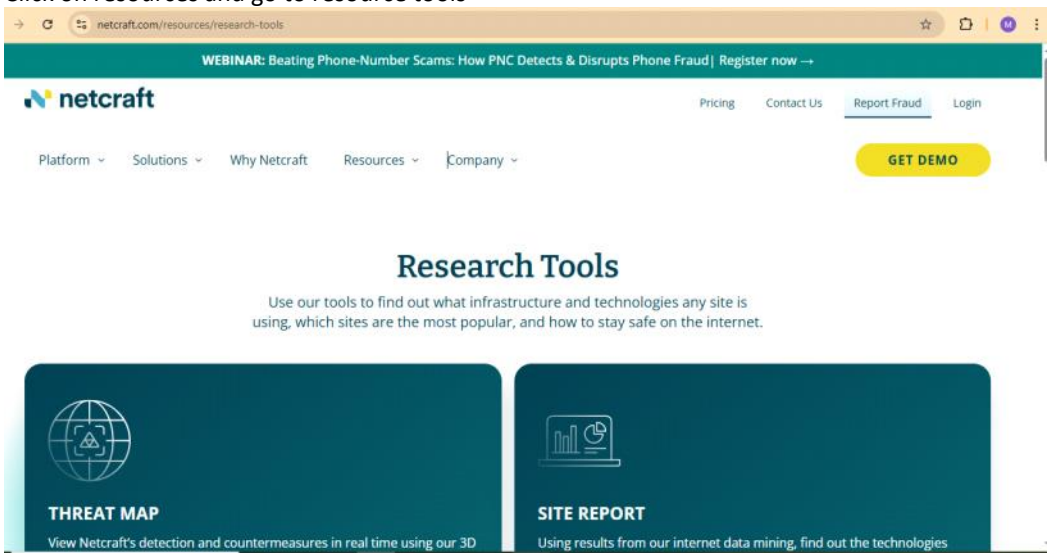
Click share button
Now open windows powershell



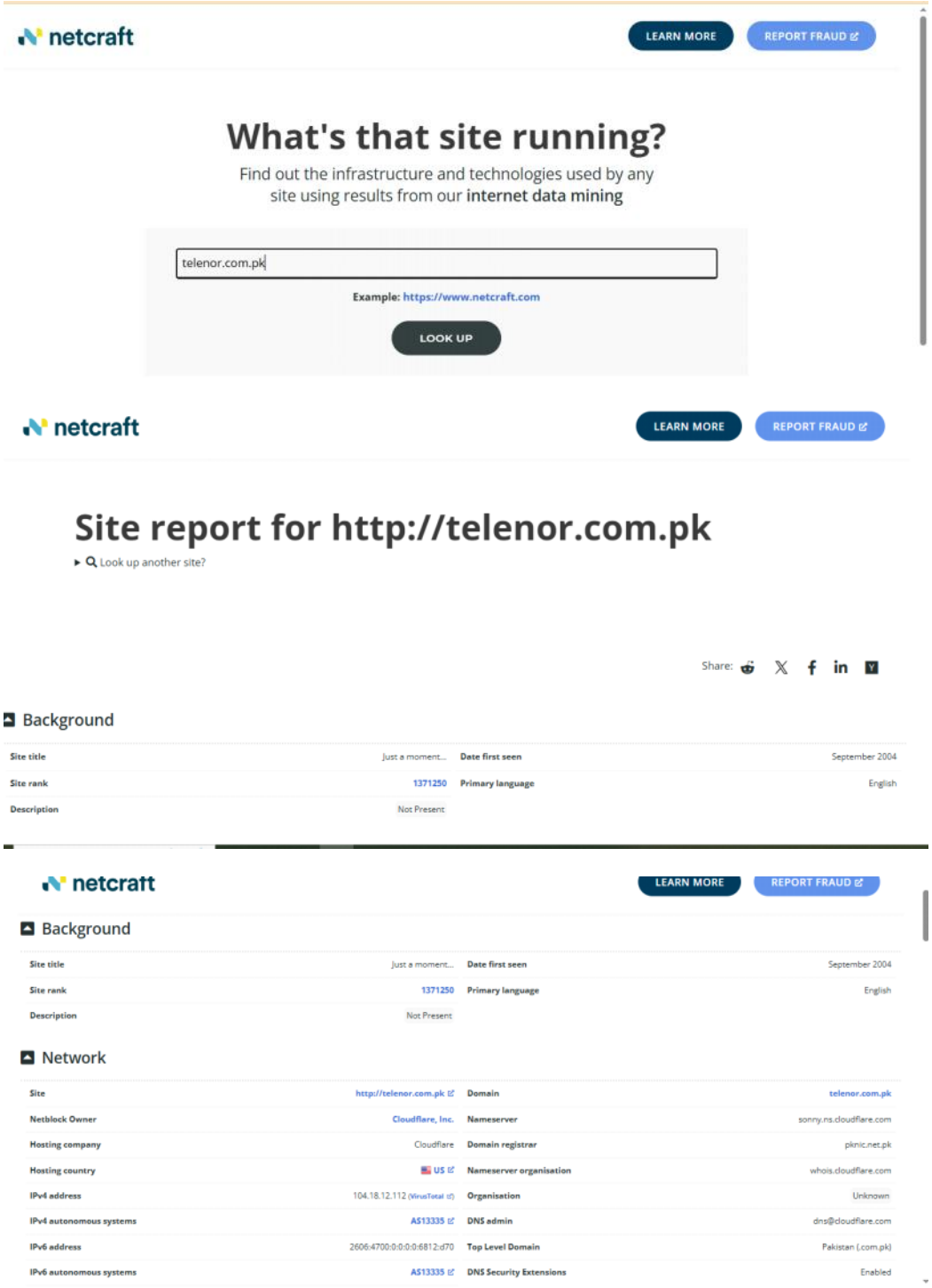
netcraft- resource tool - site report- telenor.com
copy ipv4
ip2 location geop api
paste ip
city
district
zipcode
longitude ,latitude copy
google map
paste longitude and latitude numerical values
see their server location



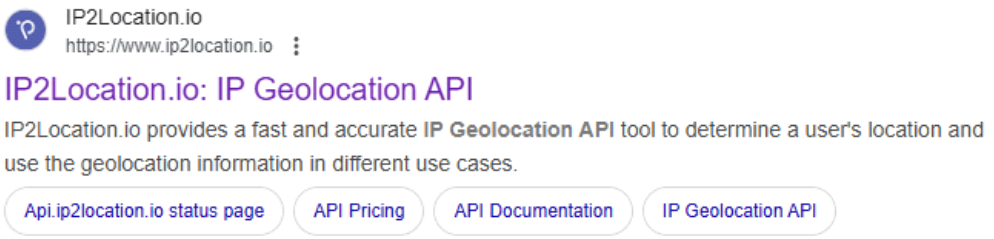
Click on resources and go to resource tools

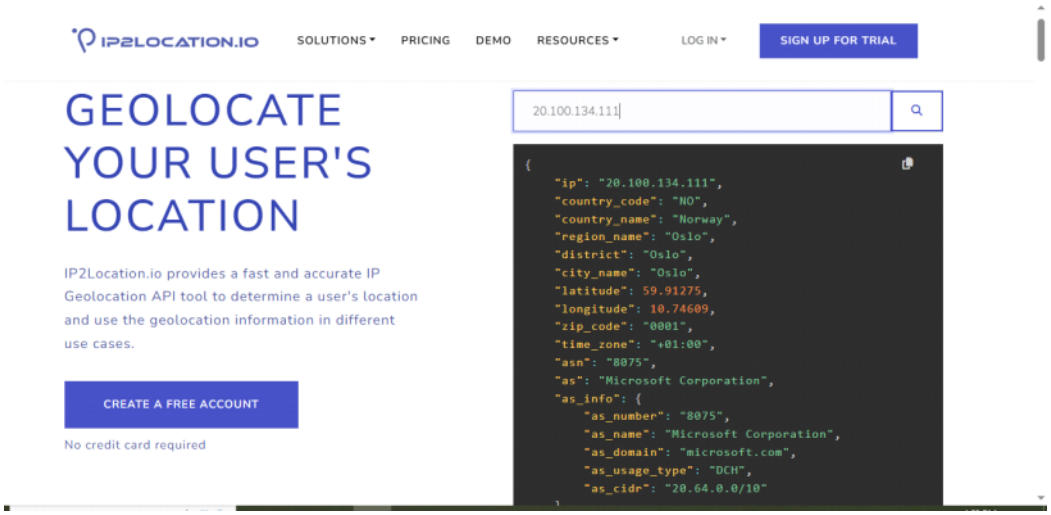


Click on site report



Copy ipv4 address and open website ip2location geop api





Copy latitude and longitude and open google map then paste the mon search bar and see the location of the server

